

Grade 7 Mathematics Complete Resource Kit

Quick Start Guide (10 Minutes to Begin)

Welcome! This kit gives you everything to support your 7th grader's math learning at home.

In your first 10 minutes:

1. **Scan Section 1** (Core Topics Overview) to see what Grade 7 covers
2. **Print the Diagnostic** (Section 2, pages marked "DIAGNOSTIC")
3. **Have your child complete it** (20 minutes, calculator OK)
4. **Score it using the rubric** provided—it tells you exactly where to start
5. **Jump to the recommended topic** and begin with Lesson A

Daily routine: 30–45 minutes, 4 days/week

- **Days 1–2:** Work through a lesson (concept + guided practice)
- **Day 3:** Complete a practice set at the right level
- **Day 4:** Do application problems or review

Everything is here: lessons, practice problems, solutions, projects, and a 4-week schedule. Just follow the sections in order for your child's needs.

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1. Core Topics Overview

Grade 7 Mathematics Priorities

Grade 7 is a pivotal year where students deepen their understanding of rational numbers, proportional reasoning, and algebraic thinking. Here are the core areas:

Topic A: Ratios & Proportional Relationships

Plain-language explanation:

Students learn to recognize and work with relationships where two quantities change at a constant rate (e.g., miles per hour, price per item). They solve problems involving unit rates, percent, and scale drawings.

Learning objectives:

- Compute unit rates (including complex fractions like $\frac{1}{2}$ mile in $\frac{1}{4}$ hour)
- Solve percent problems (tax, discounts, tips, markup)
- Recognize and represent proportional relationships with tables, graphs, and equations ($y = kx$)
- Use scale drawings and models to find actual distances and dimensions

Prerequisites to review:

- Equivalent fractions and simplifying ratios (Grade 6)
- Basic percent (finding 10%, 50%, etc.)
- Understanding of multiplication as scaling

Typical misconceptions:

- Confusing "per" with division (e.g., thinking miles per hour means miles $\div$ time, but forgetting to interpret the result)

- Adding instead of multiplying in scale problems (e.g., if scale is 1:50, thinking 3 cm on map = 53 cm actual)
 - Misapplying percent (e.g., taking 20% off twice is not 40% off total)
-

Topic B: Operations with Rational Numbers

Plain-language explanation:

Rational numbers include all integers, fractions, and decimals (positive and negative). Students master adding, subtracting, multiplying, and dividing these numbers fluently.

Learning objectives:

- Add, subtract, multiply, and divide integers (including negative numbers)
- Perform all four operations with fractions and mixed numbers
- Convert between fractions, decimals, and percents flexibly
- Apply order of operations with rational numbers
- Solve real-world problems involving rational number operations

Prerequisites to review:

- Whole number operations and order of operations (Grade 5–6)
- Fraction basics (equivalence, simplification, common denominators)
- Understanding the number line (positive and negative placement)

Typical misconceptions:

- "Two negatives make a positive" applied incorrectly to addition (e.g., $-3 + (-5) = 8$)
 - Confusion with subtraction: $5 - (-3)$ versus $5 + (-3)$
 - Dividing fractions: inverting the wrong fraction or forgetting to invert
 - Believing that multiplying always makes bigger (not true for fractions or negatives)
-

Topic C: Expressions, Equations & Inequalities

Plain-language explanation:

Students translate real-world situations into algebraic expressions and equations, then solve for unknown values. They also work with inequalities (e.g., $x > 5$) and understand their solutions.

Learning objectives:

- Simplify algebraic expressions (combine like terms, distributive property)
- Solve two-step equations (e.g., $3x + 5 = 20$)
- Solve two-step inequalities and graph solutions on a number line
- Write equations from word problems and check solutions
- Understand that expressions can be rewritten in equivalent forms

Prerequisites to review:

- Order of operations (PEMDAS)
- One-step equations (Grade 6)
- Substitution and evaluating expressions

Typical misconceptions:

- Forgetting to perform the same operation on both sides of an equation
- Distributing incorrectly: $3(x + 2) = 3x + 2$ instead of $3x + 6$
- Flipping inequality signs without dividing/multiplying by a negative
- Confusing "solve for x " with "evaluate the expression"

Topic D: Geometry (Area, Surface Area, Volume; Angles; Scale Drawings)**Plain-language explanation:**

Students calculate the area of two-dimensional shapes (including circles), surface area and volume of three-dimensional figures (prisms, pyramids, cylinders), work with angles (complementary, supplementary, vertical), and use scale drawings.

Learning objectives:

- Find circumference and area of circles using π
- Calculate surface area of prisms and pyramids by summing face areas

- Find volume of prisms, cylinders, and pyramids using formulas
- Solve problems involving angle relationships (complementary, supplementary, vertical, angles in triangles)
- Use scale factors to solve real-world problems with drawings and models

Prerequisites to review:

- Area formulas for rectangles, triangles, parallelograms (Grade 6)
- Understanding of 3D shapes and their properties
- Basic angle vocabulary (acute, obtuse, right)

Typical misconceptions:

- Confusing area and perimeter (or surface area and volume)
 - Forgetting to use radius (not diameter) in circle area formula: $A = \pi r^2$
 - Thinking surface area is just base area $\times$ height
 - Mixing up the volume formulas (e.g., cylinder vs. cone)
 - Incorrect application of scale: multiplying lengths instead of squaring for area
-

Topic E: Statistics & Probability**Plain-language explanation:**

Students learn to analyze data sets using measures of center (mean, median, mode) and measures of spread (range, interquartile range). They understand sampling, make inferences, and solve simple probability problems.

Learning objectives:

- Calculate and interpret mean, median, mode, and range
- Understand and use random sampling to draw inferences about populations
- Compare two data sets using measures of center and variability
- Find probabilities of simple events and compound events (using lists, tree diagrams, tables)
- Distinguish between theoretical and experimental probability

Prerequisites to review:

- Reading and creating bar graphs, line plots, histograms (Grade 6)
- Basic fraction and decimal understanding (for probability)

Typical misconceptions:

- Believing that mean is always the "best" measure (ignoring outliers)
 - Confusing range with interquartile range
 - Thinking probability of 0.5 means "it will happen exactly half the time in any trial"
 - Adding probabilities when they should multiply (independent events)
 - Overgeneralizing from small samples
-

2. Diagnostic & Placement

Diagnostic Assessment (20 minutes, calculator allowed)**Instructions for Parents:**

Give your child a quiet space, a calculator, pencil, and scratch paper. Set a timer for 20 minutes. Tell them to try every problem but not to stress—this helps us know where to start.

Instructions for Student:

Do your best on each problem. Show your work. If you're stuck, make your best guess and move on. This is not graded—it just helps us plan your learning.

DIAGNOSTIC TEST – GRADE 7 MATH

Name: _____ **Date:** _____

1. What is the unit rate?

A car travels 135 miles in 3 hours. How many miles per hour?

2. Solve:

$$-8 + 5 = ?$$

3. Solve for x :

$$2x - 7 = 11$$

4. A shirt costs \$25. It is on sale for 20% off. What is the sale price?

5. Simplify:

$$3(x + 4) - 2x$$

6. What is the area of a circle with radius 6 cm? (Use $\pi \approx 3.14$)

7. Find the volume of a rectangular prism with length 5 cm, width 3 cm, height 4 cm.

8. Multiply:

$$\frac{2}{3} \times \frac{5}{8} = ?$$

9. The data set is: 4, 7, 7, 10, 12. What is the median?

10. If you flip a fair coin twice, what is the probability of getting two heads?

11. Solve the inequality and graph on a number line:

$$x + 5 < 12$$

12. A scale drawing has a scale of 1 cm : 4 m. If a room is 3 cm long on the drawing, what is the actual length?

Diagnostic Answer Key & Rubric

Problem	Answer	Topic	Points
1	45 mph	Ratios & Proportional Relationships	1
2	−3	Operations with Rational Numbers	1
3	$x = 9$	Expressions, Equations & Inequalities	1
4	\$20	Ratios & Proportional Relationships	1
5	$x + 12$	Expressions, Equations & Inequalities	1
6	113.04 cm ² (accept 113–114)	Geometry	1
7	60 cm ³	Geometry	1
8	$\frac{10}{24} = \frac{5}{12}$	Operations with Rational Numbers	1
9	7	Statistics & Probability	1
10	$\frac{1}{4}$ or 0.25	Statistics & Probability	1
11	$x < 7$ (with correct number line)	Expressions, Equations & Inequalities	1
12	12 m	Geometry	1

Total: 12 points

Scoring & Placement Guide

Count correct answers by topic:

Ratios & Proportional Relationships (Problems 1, 4):

- 0–1 correct → **Start with Topic A lessons + Foundation practice**
- 2 correct → **Brief review, then On-Level practice**

Operations with Rational Numbers (Problems 2, 8):

- 0–1 correct → **Start with Topic B lessons + Foundation practice**
- 2 correct → **Brief review, then On-Level practice**

Expressions, Equations & Inequalities (Problems 3, 5, 11):

- 0–1 correct → **Start with Topic C lessons + Foundation practice**
- 2 correct → **Review one-step equations, then proceed to two-step**
- 3 correct → **On-Level practice, focus on word problems**

Geometry (Problems 6, 7, 12):

- 0–1 correct → **Start with Topic D lessons + Foundation practice**
- 2 correct → **Review formulas, then On-Level practice**
- 3 correct → **On-Level practice with application problems**

Statistics & Probability (Problems 9, 10):

- 0 correct → **Start with Topic E lessons + Foundation practice**
- 1 correct → **Quick review of definitions, then On-Level practice**
- 2 correct → **On-Level practice**

Diagnostic Summary Report Template**Student:** _____ **Date:** _____**Overall Score:** _____ / 12**Strengths:**

(List topics with 2–3 correct)

Areas for Growth:

(List topics with 0–1 correct)

Recommended Starting Points:

1. **Priority Topic:** _____ (Start here—lowest score)
2. **Next Topic:** _____
3. **Review Topic:** _____ (Scored 1–2, needs reinforcement)

Suggested Weekly Plan:

- Week 1: Focus on Priority Topic (4 days)
- Week 2: Priority Topic application + Next Topic introduction
- Week 3: Next Topic + Review Topic
- Week 4: Mixed review + Mastery Check

Parent Notes:

(Space to write observations, student's confidence level, work habits, etc.)

3. Lesson Plan Templates

For each core topic, three lessons follow the sequence: **Concept Introduction** → **Guided Practice** → **Application**. Each lesson is designed for one 30–45 minute session.

TOPIC A: Ratios & Proportional Relationships

Lesson A1: Understanding Unit Rates

Essential Vocabulary:

- **Unit rate:** a ratio comparing a quantity to 1 unit of another quantity (e.g., miles per 1 hour)
- **Rate:** a ratio comparing two different units (e.g., 120 miles in 3 hours)

"I can..." goals:

- I can find a unit rate from a given ratio
- I can solve problems involving rates using division

Concept Introduction (15 minutes):

Intuitive Explanation:

A unit rate answers "how much per one?" For example, if 3 apples cost \$6, the unit rate is \$2 per apple (divide $\$6 \div 3 \text{ apples} = \$2/\text{apple}$).

Visual:

3 apples → \$6
 1 apple → $\$6 \div 3 = \2

Think of it this way: You're figuring out the "price tag" for just one item.

Example 1:

A car travels 240 miles in 4 hours. What is the unit rate (miles per hour)?

$$\text{Unit rate} = \frac{240 \text{ miles}}{4 \text{ hours}} = 60 \text{ miles per hour}$$

Example 2 (with fractions):

You walk $\frac{1}{2}$ mile in $\frac{1}{4}$ hour. What is your walking rate in miles per hour?

$$\text{Unit rate} = \frac{\frac{1}{2} \text{ mile}}{\frac{1}{4} \text{ hour}} = \frac{1}{2} \div \frac{1}{4} = \frac{1}{2} \times \frac{4}{1} = 2 \text{ miles per hour}$$

Key Idea: Divide to find "per 1."

Guided Practice (20 minutes):

Work through these together. Student does the math, parent guides reasoning.

Problem 1:

A printer prints 180 pages in 6 minutes. How many pages per minute?

Hint: Divide total pages by total minutes.

Solution:

$$\frac{180 \text{ pages}}{6 \text{ minutes}} = 30 \text{ pages per minute}$$

Problem 2:

You earn \$48 for 8 hours of work. What is your hourly rate?

Solution:

$$\frac{48 \text{ dollars}}{8 \text{ hours}} = 6 \text{ dollars per hour}$$

Problem 3:

A recipe uses $\frac{3}{4}$ cup of sugar for $\frac{1}{2}$ batch. How much sugar per batch?

Solution:

$$\frac{\frac{3}{4} \text{ cup}}{\frac{1}{2} \text{ batch}} = \frac{3}{4} \div \frac{1}{2} = \frac{3}{4} \times 2 = \frac{6}{4} = \frac{3}{2} = 1.5 \text{ cups per batch}$$

Quick Check (5 minutes):

1. 150 words typed in 5 minutes. Words per minute?

Answer: 30 words/min

2. $\frac{2}{3}$ gallon of paint covers $\frac{1}{6}$ wall. Gallons per wall? *Answer: 4 gallons/wall*

3. \$21 for 3 pounds of cheese. Cost per pound?

Answer: \$7/pound

Extension Idea:

Find unit rates for grocery items at home. Which brand has a better unit price?

Lesson A2: Solving Percent Problems**Essential Vocabulary:**

- **Percent:** per hundred (e.g., 25% means 25 out of 100)
- **Discount:** reduction in price
- **Markup:** increase in price
- **Tax/Tip:** additional amount added to a bill

"I can..." goals:

- I can find a percent of a number
- I can calculate sale prices, tax, and tips
- I can solve percent increase and decrease problems

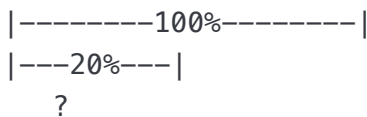
Concept Introduction (15 minutes):

Intuitive Explanation:

Percent problems ask "what is this fraction of 100, applied to my number?" For example, 20% of 50 means $\frac{20}{100} \times 50 = 0.20 \times 50 = 10$.

Visual Method (Percent Bar):

Draw a bar representing the whole (100%).



If $100\% = \$50$, then $20\% = \$50 \times 0.20 = \10 .

Three Key Percent Problems:

1. Finding a percent of a number:

What is 30% of 80?

$$0.30 \times 80 = 24$$

2. Finding sale price after discount:

Original price: \$60. Discount: 25% off.

Discount amount: $\$60 \times 0.25 = \15

Sale price: $\$60 - \$15 = \$45$

OR shortcut: $\$60 \times 0.75 = \45 (since $100\% - 25\% = 75\%$)

3. Finding total after tax or tip:

Meal: \$40. Tax: 8%.

Tax amount: $\$40 \times 0.08 = \3.20

Total: $\$40 + \$3.20 = \$43.20$

OR shortcut: $\$40 \times 1.08 = \43.20

Guided Practice (20 minutes):

Problem 1:

A bicycle costs \$250. It's on sale for 15% off. What is the sale price?

Reasoning:

- Find 15% of \$250: $\$250 \times 0.15 = \37.50
- Subtract from original: $\$250 - \$37.50 = \$212.50$
OR $\$250 \times 0.85 = \212.50

Problem 2:

You leave a 20% tip on a \$32 restaurant bill. How much is the tip? What's the total?

Reasoning:

- Tip: $\$32 \times 0.20 = \6.40
- Total: $\$32 + \$6.40 = \$38.40$
OR $\$32 \times 1.20 = \38.40

Problem 3:

A store marks up a shirt by 40%. The original cost to the store was \$25. What is the selling price?

Reasoning:

- Markup: $\$25 \times 0.40 = \10
- Selling price: $\$25 + \$10 = \$35$
OR $\$25 \times 1.40 = \35

Quick Check (5 minutes):

1. What is 12% of 150?

Answer: 18

2. A \$90 jacket is 30% off. Sale price?

Answer: \$63

3. You pay 7% sales tax on a \$50 item. Total cost?

Answer: \$53.50

Extension Idea:

Calculate your total cost for 3 items with different discounts and add tax. Compare buying strategies (coupon vs. sale).

Lesson A3: Proportional Relationships & Graphs

Essential Vocabulary:

- **Proportional relationship:** a relationship where the ratio of two quantities is constant (e.g., $y = kx$ where k is the constant of proportionality)
- **Constant of proportionality:** the constant ratio k in a proportional relationship

"I can..." goals:

- I can recognize proportional relationships in tables and graphs
- I can find the constant of proportionality
- I can write equations for proportional relationships ($y = kx$)

Concept Introduction (15 minutes):

Intuitive Explanation:

In a proportional relationship, if one quantity doubles, the other doubles too. The graph is a straight line through the origin (0, 0).

Example:

You earn \$12 per hour. If you work 1 hour, you earn \$12. Work 2 hours, earn \$24. Work 3 hours, earn \$36.

Hours (x)	Earnings (y)
1	12
2	24
3	36

Notice: $\frac{y}{x} = \frac{12}{1} = \frac{24}{2} = \frac{36}{3} = 12$ (constant ratio).

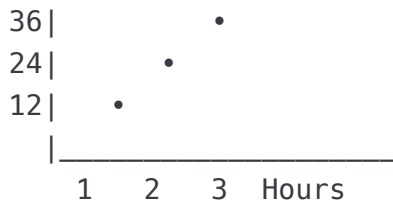
The constant of proportionality is $k = 12$.

Equation: $y = 12x$ (earnings = \$12 $\times$ hours)

Graph: A straight line through (0,0) with slope 12.

Visual:





Key Test for Proportionality:

1. Does the graph pass through (0, 0)?
2. Is the ratio $\frac{y}{x}$ the same for all points?

Guided Practice (20 minutes):

Problem 1:

The table shows the cost of apples. Is this proportional? If so, find k and write the equation.

Pounds (x)	Cost (y)
2	5
4	10
6	15

Reasoning:

- Check ratios: $\frac{5}{2} = 2.5$, $\frac{10}{4} = 2.5$, $\frac{15}{6} = 2.5$ ✓ Constant
- $k = 2.5$
- Equation: $y = 2.5x$
- Yes, it's proportional.

Problem 2:

Is this table proportional?

x	y
1	3
2	6
3	10

Reasoning:

- Check ratios: $\frac{3}{1} = 3$, $\frac{6}{2} = 3$, $\frac{10}{3} \approx 3.33$ ✗ Not constant

- Not proportional.

Problem 3:

A car uses 2 gallons of gas to travel 60 miles. Write a proportional equation for miles (y) in terms of gallons (x).

Reasoning:

- $k = \frac{60}{2} = 30$ miles per gallon
- Equation: $y = 30x$

Quick Check (5 minutes):

1. If $y = 7x$, what is the constant of proportionality? *Answer: 7*
2. A proportional graph passes through (4, 20). Write the equation. *Answer: $y = 5x$ (since $k = \frac{20}{4} = 5$)*
3. Is $y = 3x + 2$ proportional? *Answer: No (doesn't pass through origin)*

Extension Idea:

Graph your own proportional relationship (e.g., pages read per day) and predict how many pages in 10 days.

TOPIC B: Operations with Rational Numbers

Lesson B1: Adding & Subtracting Integers

Essential Vocabulary:

- **Integer:** whole numbers and their opposites (e.g., -3, 0, 5)
- **Absolute value:** distance from zero (always positive, denoted $|-5| = 5$)
- **Opposite:** the number that, when added to the original, gives zero (e.g., opposite of 3 is -3)

"I can..." goals:

- I can add and subtract positive and negative integers
- I can use a number line to visualize integer operations

Concept Introduction (15 minutes):

Intuitive Explanation:

Think of a number line as a thermometer or a bank account. Positive numbers are gains, negative numbers are losses.

Number Line Visual:

-5 -4 -3 -2 -1 0 1 2 3 4 5

Adding Integers:

- **Same signs:** Add the absolute values, keep the sign.
 $-3 + (-5) = -8$ (moving left 3, then left 5 more)
 $4 + 6 = 10$
- **Different signs:** Subtract the absolute values, take the sign of the larger absolute value.
 $-7 + 4 = -3$ ($7 > 4$, so answer is negative)
 $8 + (-5) = 3$ ($8 > 5$, so answer is positive)

Subtracting Integers:

Key Rule: Subtracting is the same as adding the opposite.

$$5 - 8 = 5 + (-8) = -3$$

$$-3 - 7 = -3 + (-7) = -10$$

$$-6 - (-4) = -6 + 4 = -2$$

Guided Practice (20 minutes):**Problem 1:**

Calculate: $-12 + 8$

Reasoning:

- Different signs: $12 - 8 = 4$
- Larger absolute value is 12 (negative), so answer is negative.
- Answer: -4

Problem 2:

Calculate: $7 - (-3)$

Reasoning:

- Rewrite as adding the opposite: $7 + 3 = 10$

- Answer: 10

Problem 3:

Calculate: $-5 + (-9)$

Reasoning:

- Same signs (both negative): $5 + 9 = 14$, keep negative sign.
- Answer: -14

Quick Check (5 minutes):

1. $-8 + 15 = ?$ *Answer: 7*
2. $6 - 11 = ?$ *Answer: -5*
3. $-4 - (-10) = ?$ *Answer: 6*

Extension Idea:

Track daily temperature changes using integers. If it's 3°F and drops 8° , what's the new temperature?

Lesson B2: Multiplying & Dividing Integers**Essential Vocabulary:**

- **Product:** result of multiplication
- **Quotient:** result of division

"I can..." goals:

- I can multiply and divide integers using sign rules
- I can explain why the product of two negatives is positive

Concept Introduction (15 minutes):**Sign Rules (easy to remember):****Multiplication:**

- **Same signs** $\rightarrow$ **Positive**
 $3 \times 4 = 12$

$$(-3) \times (-4) = 12$$

- **Different signs → Negative**

$$3 \times (-4) = -12$$

$$(-3) \times 4 = -12$$

Division (same rules):

- **Same signs → Positive**

$$12 \div 3 = 4$$

$$(-12) \div (-3) = 4$$

- **Different signs → Negative**

$$12 \div (-3) = -4$$

$$(-12) \div 3 = -4$$

Why does negative $\times$ negative = positive?

Think of $-3 \times (-4)$ as "the opposite of 3 groups of -4."

Three groups of -4 is -12. The opposite of -12 is +12.

Guided Practice (20 minutes):

Problem 1:

Calculate: $(-7) \times 5$

Reasoning:

- Different signs → negative
- $7 \times 5 = 35$
- Answer: -35

Problem 2:

Calculate: $(-6) \times (-8)$

Reasoning:

- Same signs (both negative) → positive
- $6 \times 8 = 48$
- Answer: 48

Problem 3:

Calculate: $(-20) \div (-4)$

Reasoning:

- Same signs $\rightarrow$ positive
- $20 \div 4 = 5$
- Answer: 5

Quick Check (5 minutes):

1. $(-9) \times 3 = ?$ *Answer: -27*
2. $(-15) \div 5 = ?$ *Answer: -3*
3. $(-4) \times (-6) = ?$ *Answer: 24*

Extension Idea:

Create a "sign chart" poster: draw four quadrants for ++, +-, -+, -- and mark positive or negative results.

Lesson B3: Operations with Fractions & Decimals**Essential Vocabulary:**

- **Common denominator:** same denominator for two or more fractions, needed for adding/subtracting
- **Reciprocal:** the "flip" of a fraction (e.g., reciprocal of $\frac{2}{3}$ is $\frac{3}{2}$)

"I can..." goals:

- I can add and subtract fractions with different denominators
- I can multiply and divide fractions fluently
- I can convert between fractions and decimals

Concept Introduction (15 minutes):**Adding/Subtracting Fractions:**

Need common denominators.

Example: $\frac{1}{4} + \frac{1}{6}$

Find LCD (Least Common Denominator): 12

$$\frac{1}{4} = \frac{3}{12}, \quad \frac{1}{6} = \frac{2}{12}$$

$$\frac{3}{12} + \frac{2}{12} = \frac{5}{12}$$

Multiplying Fractions:

Multiply straight across. No common denominator needed.

$$\frac{2}{3} \times \frac{4}{5} = \frac{2 \times 4}{3 \times 5} = \frac{8}{15}$$

Dividing Fractions:

Multiply by the reciprocal ("Keep, Change, Flip").

$$\frac{2}{3} \div \frac{4}{5} = \frac{2}{3} \times \frac{5}{4} = \frac{10}{12} = \frac{5}{6}$$

Decimals:

Line up decimal points for adding/subtracting.

For multiplying, ignore decimals, multiply, then count total decimal places.

For dividing, move decimal in divisor to make it a whole number, move same in

Continue

dividend.

Guided Practice (20 minutes):

Problem 1:

Calculate: $\frac{3}{5} + \frac{1}{4}$

Reasoning:

- LCD = 20
- $\frac{3}{5} = \frac{12}{20}, \frac{1}{4} = \frac{5}{20}$
- $\frac{12}{20} + \frac{5}{20} = \frac{17}{20}$

Problem 2:

Calculate: $\frac{5}{8} \times \frac{2}{3}$

Reasoning:

- $\frac{5 \times 2}{8 \times 3} = \frac{10}{24} = \frac{5}{12}$

Problem 3:

Calculate: $\frac{7}{10} \div \frac{2}{5}$

Reasoning:

- Multiply by reciprocal: $\frac{7}{10} \times \frac{5}{2} = \frac{35}{20} = \frac{7}{4} = 1 \frac{3}{4}$

Quick Check (5 minutes):

- $\frac{2}{3} - \frac{1}{6} = ?$ *Answer: $\frac{1}{2}$ *
- $\frac{3}{4} \times \frac{2}{9} = ?$ *Answer: $\frac{1}{6}$ *
- $0.6 + 0.45 = ?$ *Answer: 1.05 *

Extension Idea:

Measure ingredients for a recipe in fractions, then double the recipe (multiply all fractions by 2).

TOPIC C: Expressions, Equations & Inequalities

Lesson C1: Simplifying Expressions

Essential Vocabulary:

- Expression:** a math phrase (no equal sign) like $3x + 5$
- Like terms:** terms with the same variable part (e.g., $2x$ and $5x$)
- Distributive property:** $a(b + c) = ab + ac$

"I can..." goals:

- I can combine like terms
- I can use the distributive property to expand expressions
- I can simplify multi-step expressions

Concept Introduction (15 minutes):

Combining Like Terms:

Only add/subtract terms with the exact same variable.

$$5x + 3x = 8x \quad (\text{like terms})$$

$$5x + 3y \quad (\text{cannot combine—different variables})$$

Distributive Property:

Multiply the outside number by each term inside the parentheses.

$$3(x + 4) = 3x + 12$$

$$-2(3x - 5) = -6x + 10$$

Multi-Step Example:

$$2(3x + 1) - 4x$$

$$= 6x + 2 - 4x \quad (\text{distribute})$$

$$= 2x + 2 \quad (\text{combine like terms})$$

Guided Practice (20 minutes):**Problem 1:**

Simplify: $7a + 3a - 2a$

Reasoning:

- All like terms: $7 + 3 - 2 = 8$
- Answer: $8a$

Problem 2:

Simplify: $4(x + 3) + 2x$

Reasoning:

- Distribute: $4x + 12 + 2x$
- Combine: $6x + 12$

Problem 3:

Simplify: $5(2y - 1) - 3(y + 2)$

Reasoning:

- Distribute: $10y - 5 - 3y - 6$
- Combine: $7y - 11$

Quick Check (5 minutes):

1. $6b + 2b = ?$ *Answer: $8b$ *
2. $3(x + 5) = ?$ *Answer: $3x + 15$ *
3. $2(3m - 4) + m = ?$ *Answer: $7m - 8$ *

Extension Idea:

Write an expression for the perimeter of a rectangle with sides $x + 3$ and $2x$, then simplify.

Lesson C2: Solving Two-Step Equations**Essential Vocabulary:**

- **Equation:** a math sentence with an equal sign (e.g., $2x + 3 = 11$)
- **Solution:** the value of the variable that makes the equation true
- **Inverse operations:** operations that undo each other (+ and -, $\times$ and $\div$)

"I can..." goals:

- I can solve two-step equations using inverse operations
- I can check my solutions by substitution

Concept Introduction (15 minutes):

Strategy: Undo operations in reverse order (opposite of order of operations).

Example: Solve $2x + 5 = 13$

Steps:

1. Subtract 5 from both sides (undo the +5):

$$2x + 5 - 5 = 13 - 5$$

$$2x = 8$$

2. Divide both sides by 2 (undo the $\times 2$):

$$\frac{2x}{2} = \frac{8}{2}$$

$$x = 4$$

Check: Substitute $x = 4$ into original:

$$2(4) + 5 = 8 + 5 = 13 \quad \checkmark$$

Visual (Balance Scale):

$$\begin{array}{lcl} 2x + 5 & = & 13 \\ \text{(take away 5 from both sides)} & & \\ 2x & = & 8 \\ \text{(divide both by 2)} & & \\ x & = & 4 \end{array}$$

Guided Practice (20 minutes):

Problem 1:

Solve: $3x - 7 = 14$

Reasoning:

- Add 7 to both sides: $3x = 21$
- Divide by 3: $x = 7$
- Check: $3(7) - 7 = 21 - 7 = 14 \quad \checkmark$

Problem 2:

Solve: $\frac{x}{4} + 2 = 5$

Reasoning:

- Subtract 2: $\frac{x}{4} = 3$
- Multiply by 4: $x = 12$
- Check: $\frac{12}{4} + 2 = 3 + 2 = 5 \checkmark$

Problem 3:

Solve: $-2x + 6 = 0$

Reasoning:

- Subtract 6: $-2x = -6$
- Divide by -2: $x = 3$
- Check: $-2(3) + 6 = -6 + 6 = 0 \checkmark$

Quick Check (5 minutes):

1. Solve: $5x + 3 = 18$ *Answer: $x = 3$ *
2. Solve: $\frac{x}{2} - 4 = 1$ *Answer: $x = 10$ *
3. Solve: $7 - 3x = -2$ *Answer: $x = 3$ *

Extension Idea:

Create your own two-step equations and trade with a friend to solve.

Lesson C3: Writing & Solving Equations from Word Problems**Essential Vocabulary:**

- **Variable:** a letter representing an unknown number
- **Translate:** change words into math symbols

"I can..." goals:

- I can identify the variable in a word problem
- I can write an equation from a word problem
- I can solve the equation and answer the question in context

Concept Introduction (15 minutes):**Strategy—Four Steps:**

1. **Identify** what you're solving for (define the variable)
2. **Translate** words to math
3. **Solve** the equation
4. **Answer** the question in a complete sentence

Key Phrases:

Word Phrase	Math Symbol
more than, sum, total, increased by	+
less than, difference, decreased by	-
times, product, of (with fractions/percent)	$\times$
divided by, quotient, per	$\div$
is, equals, gives, results in	=

Example:

"Five more than twice a number is 17. Find the number."

1. Let x = the number
2. Translate: "twice a number" = $2x$; "five more than" = $2x + 5$; "is 17" = $= 17$
Equation: $2x + 5 = 17$
3. Solve: $2x = 12$, so $x = 6$
4. Answer: The number is 6.

Guided Practice (20 minutes):

Problem 1:

"Maria bought a shirt for \$18 and some pants. She spent \$42 total. How much were the pants?"

Reasoning:

1. Let p = cost of pants
2. Equation: $18 + p = 42$
3. Solve: $p = 24$
4. Answer: The pants cost \$24.

Problem 2:

"Three times a number, decreased by 8, equals 19. What is the number?"

Reasoning:

1. Let n = the number
2. Equation: $3n - 8 = 19$
3. Solve: $3n = 27, n = 9$
4. Answer: The number is 9.

Problem 3:

"A taxi charges \$3 plus \$2 per mile. You pay \$15. How many miles did you travel?"

Reasoning:

1. Let m = miles
2. Equation: $3 + 2m = 15$
3. Solve: $2m = 12, m = 6$
4. Answer: You traveled 6 miles.

Quick Check (5 minutes):

1. "Seven less than a number is 10." Write and solve the equation. *Answer: $x - 7 = 10, x = 17$ *
2. "Four times a number, plus 5, equals 21." Find the number. *Answer: $4x + 5 = 21, x = 4$ *
3. "You divide a number by 3 and add 2 to get 8." Find the number. *Answer: $\frac{x}{3} + 2 = 8, x = 18$ *

Extension Idea:

Write a word problem for a friend based on something from your day (allowance, shopping, games).

TOPIC D: Geometry**Lesson D1: Circumference & Area of Circles****Essential Vocabulary:**

- **Radius (r)** : distance from center to edge

- **Diameter (d)** : distance across circle through center ($d = 2r$)
- **Pi (π)** : the ratio of circumference to diameter, approximately 3.14
- **Circumference (C)** : perimeter of a circle
- **Area (A)** : space inside a circle

"I can..." goals:

- I can find the circumference of a circle using $C = \pi d$ or $C = 2\pi r$
- I can find the area of a circle using $A = \pi r^2$
- I can use radius or diameter correctly in formulas

Concept Introduction (15 minutes):**Formulas:**

$$C = \pi d \quad \text{or} \quad C = 2\pi r$$

$$A = \pi r^2$$

Remember: Circumference is linear (distance around), Area is square units (space inside).

Example 1:

A circle has radius 5 cm. Find circumference and area. (Use $\pi \approx 3.14$)

Circumference:

$$C = 2\pi r = 2 \times 3.14 \times 5 = 31.4 \text{ cm}$$

Area:

$$A = \pi r^2 = 3.14 \times 5^2 = 3.14 \times 25 = 78.5 \text{ cm}^2$$

Example 2:

A circle has diameter 10 inches. Find the area.

Solution:

- Radius = diameter $\div$ 2 = 5 inches
- $A = \pi r^2 = 3.14 \times 5^2 = 78.5 \text{ in}^2$

Guided Practice (20 minutes):

Problem 1:

A pizza has a radius of 8 inches. What is its circumference?

Reasoning:

$$C = 2\pi r = 2 \times 3.14 \times 8 = 50.24 \text{ inches}$$

Problem 2:

A circular garden has a diameter of 12 feet. What is its area?

Reasoning:

- Radius = 6 feet
- $A = \pi r^2 = 3.14 \times 6^2 = 3.14 \times 36 = 113.04 \text{ ft}^2$

Problem 3:

A wheel has circumference 62.8 cm. What is its radius? (Use $\pi \approx 3.14$)

Reasoning:

- $C = 2\pi r$
- $62.8 = 2 \times 3.14 \times r$
- $62.8 = 6.28r$
- $r = 10 \text{ cm}$

Quick Check (5 minutes):

1. Radius = 3 m. Find circumference.

Answer: 18.84 m

2. Diameter = 14 cm. Find area.

Answer: 153.86 cm²

3. Area = 50.24 in². Find radius. ($\pi \approx 3.14$) **Answer: 4 in**

Extension Idea:

Measure circular objects at home (plates, lids). Calculate their area and circumference. Check with string!

Lesson D2: Surface Area of Prisms & Pyramids**Essential Vocabulary:**

- **Prism:** a 3D shape with two parallel, congruent bases (e.g., rectangular prism, triangular prism)
- **Pyramid:** a 3D shape with one base and triangular faces meeting at a point
- **Surface area:** total area of all faces (measured in square units)
- **Net:** a 2D pattern that folds into a 3D shape

"I can..." goals:

- I can identify the faces of a prism or pyramid
- I can calculate surface area by finding the area of each face and adding
- I can use nets to visualize surface area

Concept Introduction (15 minutes):

Strategy: Find the area of each face, then add them all together.

Rectangular Prism:

Has 6 rectangular faces.

Example: Length = 5 cm, Width = 3 cm, Height = 4 cm

Faces:

- Top and bottom: $5 \times 3 = 15 \text{ cm}^2$ each (2 faces = 30 cm^2)
- Front and back: $5 \times 4 = 20 \text{ cm}^2$ each (2 faces = 40 cm^2)
- Left and right: $3 \times 4 = 12 \text{ cm}^2$ each (2 faces = 24 cm^2)

Total surface area: $30 + 40 + 24 = 94 \text{ cm}^2$

Formula shortcut:

$$SA = 2(lw + lh + wh)$$

$$SA = 2(15 + 20 + 12) = 2(47) = 94 \text{ cm}^2$$

Triangular Prism:

Two triangular bases + three rectangular faces.

Square Pyramid:

One square base + four triangular faces.

Guided Practice (20 minutes):**Problem 1:**

Find the surface area of a rectangular prism: length = 6 in, width = 4 in, height = 3 in.

Reasoning:

- Top/bottom: $6 \times 4 = 24 \text{ in}^2$ each $\rightarrow 2 \times 24 = 48 \text{ in}^2$
- Front/back: $6 \times 3 = 18 \text{ in}^2$ each $\rightarrow 2 \times 18 = 36 \text{ in}^2$
- Sides: $4 \times 3 = 12 \text{ in}^2$ each $\rightarrow 2 \times 12 = 24 \text{ in}^2$
- Total: $48 + 36 + 24 = 108 \text{ in}^2$

Problem 2:

A square pyramid has a square base with side 5 cm and triangular faces with height 8 cm. Find surface area.

Reasoning:

- Base: $5 \times 5 = 25 \text{ cm}^2$
- Each triangular face: $\frac{1}{2} \times 5 \times 8 = 20 \text{ cm}^2$
- Four triangular faces: $4 \times 20 = 80 \text{ cm}^2$
- Total: $25 + 80 = 105 \text{ cm}^2$

Problem 3:

Draw a net for a cube with edge 4 cm, then find its surface area.

Reasoning:

- A cube has 6 square faces

- Each face: $4 \times 4 = 16 \text{ cm}^2$
- Total: $6 \times 16 = 96 \text{ cm}^2$

Quick Check (5 minutes):

1. Cube with edge 3 ft. Surface area?

Answer: 54 ft^2

2. Rectangular prism: $2 \times 3 \times 4$ m. Surface area?

Answer: 52 m^2

3. How many faces does a triangular prism have?

Answer: 5 (2 triangular, 3 rectangular)

Extension Idea:

Create a net for a box (cereal, tissue). Calculate its surface area. How much cardboard is needed?

Lesson D3: Volume of Prisms & Cylinders

Essential Vocabulary:

- **Volume:** amount of space inside a 3D shape (measured in cubic units)
- **Cylinder:** a 3D shape with two circular bases

"I can..." goals:

- I can calculate the volume of prisms using $V = Bh$ (where B = base area, h = height)
- I can calculate the volume of a cylinder using $V = \pi r^2 h$

Concept Introduction (15 minutes):

Key Formula for All Prisms:

$$V = B \times h$$

where B = area of the base, h = height

Rectangular Prism:

Base is a rectangle: $B = l \times w$

$$V = l \times w \times h$$

Example: Length = 5 cm, Width = 3 cm, Height = 4 cm

$$V = 5 \times 3 \times 4 = 60 \text{ cm}^3$$

Triangular Prism:

Base is a triangle: $B = \frac{1}{2} \times \text{base} \times \text{height of triangle}$

$$V = B \times h_{\text{prism}}$$

Cylinder:

Base is a circle: $B = \pi r^2$

$$V = \pi r^2 h$$

Example: Radius = 3 cm, Height = 10 cm

$$V = 3.14 \times 3^2 \times 10 = 3.14 \times 9 \times 10 = 282.6 \text{ cm}^3$$

Guided Practice (20 minutes):

Problem 1:

Find the volume of a rectangular prism: 8 ft × 5 ft × 3 ft.

Reasoning:

$$V = 8 \times 5 \times 3 = 120 \text{ ft}^3$$

Problem 2:

A cylinder has radius 4 inches and height 7 inches. Find the volume. (Use $\pi \approx 3.14$)

Reasoning:

$$V = \pi r^2 h = 3.14 \times 4^2 \times 7 = 3.14 \times 16 \times 7 = 351.68 \text{ in}^3$$

Problem 3:

A triangular prism has a base with triangle dimensions: base = 6 cm, height = 4 cm. The prism height is 10 cm. Find volume.

Reasoning:

- Base area: $B = \frac{1}{2} \times 6 \times 4 = 12 \text{ cm}^2$
- Volume: $V = 12 \times 10 = 120 \text{ cm}^3$

Quick Check (5 minutes):

1. Rectangular prism: $2 \times 3 \times 5$ m. Volume?

Answer: 30 m^3

2. Cylinder: radius 5 cm, height 6 cm. Volume? ($\pi \approx 3.14$) *Answer: 471 cm^3 *

3. If a box holds 100 cm^3 , what is the unit of measurement?

Answer: cubic centimeters

Extension Idea:

Calculate how much water a cylindrical water bottle holds. Compare to the label!

TOPIC E: Statistics & Probability**Lesson E1: Measures of Center & Spread****Essential Vocabulary:**

- **Mean:** the average (add all values, divide by the count)
- **Median:** the middle value when data is ordered
- **Mode:** the value that appears most often
- **Range:** difference between maximum and minimum
- **Outlier:** a value much higher or lower than the rest

"I can..." goals:

- I can calculate mean, median, mode, and range
- I can choose the best measure of center for a data set
- I can identify outliers and explain their effect

Concept Introduction (15 minutes):**Data Set Example:** 3, 5, 7, 7, 10, 12, 25**Mean:**

$$\text{Mean} = \frac{3 + 5 + 7 + 7 + 10 + 12 + 25}{7} = \frac{69}{7} \approx 9.86$$

Median:

Order the data (already ordered): 3, 5, 7, 7, 10, 12, 25

Middle value = 7

Mode:7 appears twice (most frequent) → **Mode = 7****Range:**

$$\text{Range} = 25 - 3 = 22$$

Outlier: 25 is much larger than the rest → it's an outlier. It pulls the mean up (9.86), but doesn't affect the median (7). **Median is better here.**

When to Use Which:

- **Mean:** Use when data is fairly even, no outliers
- **Median:** Use when there are outliers or skewed data
- **Mode:** Use for categorical data or to find most common

Guided Practice (20 minutes):**Problem 1:**

Find mean, median, mode, and range: 8, 12, 15, 12, 10, 9, 12

Reasoning:

- Order: 8, 9, 10, 12, 12, 12, 15
- Mean: $(8 + 9 + 10 + 12 + 12 + 12 + 15) \div 7 = 78 \div 7 \approx 11.14$
- Median: Middle value = 12
- Mode: 12 (appears 3 times)

- Range: $15 - 8 = 7$

Problem 2:

Test scores: 85, 90, 88, 92, 45. Which measure of center best represents typical performance?

Reasoning:

- Mean: $(85 + 90 + 88 + 92 + 45) \div 5 = 80$
- Median: Order: 45, 85, 88, 90, 92 $\rightarrow$ Median = 88
- The 45 is an outlier. **Median (88) is better**—it shows typical performance.

Problem 3:

Shoe sizes in a group: 7, 8, 7, 9, 7, 10. What's the mode?

Reasoning:

- 7 appears 3 times $\rightarrow$ Mode = 7

Quick Check (5 minutes):

1. Data: 4, 6, 6, 8, 10. Find the median.

Answer: 6

2. Data: 20, 22, 100, 24, 26. Which is better: mean or median? Why?

Answer: Median (100 is an outlier)

3. Data: 5, 5, 7, 9. Find the range.

Answer: 4

Extension Idea:

Collect data (family ages, daily steps). Calculate all measures. Graph them!

Lesson E2: Understanding Probability**Essential Vocabulary:**

- **Probability:** the likelihood of an event happening (from 0 to 1, or 0% to 100%)
- **Theoretical probability:** expected probability based on logic
- **Experimental probability:** probability based on actual trials
- **Sample space:** all possible outcomes

"I can..." goals:

- I can calculate theoretical probability using $P = \frac{\text{favorable outcomes}}{\text{total outcomes}}$
- I can conduct experiments and find experimental probability
- I can compare theoretical and experimental results

Concept Introduction (15 minutes):**Probability Scale:**

0	0.25	0.5	0.75	1
Impossible	Unlikely	Equally Likely	Likely	Certain

Theoretical Probability:

$$P(\text{event}) = \frac{\text{number of favorable outcomes}}{\text{total number of outcomes}}$$

Example 1: Rolling a die

What's the probability of rolling a 4?

- Favorable outcomes: 1 (just the 4)
- Total outcomes: 6 (numbers 1–6)
- $P(4) = \frac{1}{6} \approx 0.167$ or about 16.7%

Example 2: Spinner with 8 equal sections (3 red, 2 blue, 3 green)

What's $P(\text{red})$?

$$P(\text{red}) = \frac{3}{8} = 0.375 = 37.5\%$$

Experimental Probability:

Based on actual trials.

If you flip a coin 20 times and get heads 12 times:

$$P(\text{heads}) = \frac{12}{20} = 0.6 = 60\%$$

Over many trials, experimental probability should get closer to theoretical (50% for heads).

Guided Practice (20 minutes):

Problem 1:

A bag has 5 red marbles, 3 blue marbles, and 2 green marbles. What's the probability of drawing a blue marble?

Reasoning:

- Favorable: 3 blue
- Total: $5 + 3 + 2 = 10$
- $P(\text{blue}) = \frac{3}{10} = 0.3 = 30\%$

Problem 2:

You roll a die 30 times and get a 6 five times. What's the experimental probability of rolling a 6?

Reasoning:

$$P(6) = \frac{5}{30} = \frac{1}{6} \approx 0.167 = 16.7\%$$

(This matches the theoretical probability!)

Problem 3:

What's the probability of drawing an ace from a standard deck of 52 cards?

Reasoning:

- Favorable: 4 aces
- Total: 52 cards
- $P(\text{ace}) = \frac{4}{52} = \frac{1}{13} \approx 0.077 = 7.7\%$

Quick Check (5 minutes):

1. A spinner has 6 equal sections: 2 yellow, 4 red. $P(\text{yellow})$? **Answer: $\frac{1}{3}$ **

2. You flip a coin 50 times, get tails 28 times. Experimental $P(\text{tails})$? *Answer: $\frac{28}{50} = 0.56 = 56\%$ *

3. True or False: A probability of 0 means the event is certain.

Answer: False (0 means impossible)

Extension Idea:

Design your own spinner. Predict probabilities, then spin 20 times and compare results.

Lesson E3: Compound Events & Tree Diagrams

Essential Vocabulary:

- **Compound event:** two or more events happening together (e.g., flipping a coin AND rolling a die)
- **Independent events:** one event doesn't affect the other

"I can..." goals:

- I can list sample spaces for compound events
- I can use tree diagrams or tables to organize outcomes
- I can calculate probabilities of compound events

Concept Introduction (15 minutes):

Example: Flip a coin, then roll a die

Sample Space (all outcomes):

H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6

Total outcomes: $2 \times 6 = 12$

Tree Diagram:

Coin		Die		Outcome
H	-----	1	-----	H1
	-----	2	-----	H2
	-----	3	-----	H3

	-----	4	-----	H4
	-----	5	-----	H5
	-----	6	-----	H6
T	-----	1	-----	T1
	-----	2	-----	T2
	-----	3	-----	T3
	-----	4	-----	T4
	-----	5	-----	T5
	-----	6	-----	T6

Question: What's the probability of getting heads AND rolling a 3?

- Favorable: H3 (1 outcome)
- Total: 12
- $P(\text{H and } 3) = \frac{1}{12} \approx 0.083 = 8.3\%$

Multiplication Rule for Independent Events:

$$P(\text{A and B}) = P(\text{A}) \times P(\text{B})$$

Example: $P(\text{H}) = \frac{1}{2}$, $P(3) = \frac{1}{6}$

$$P(\text{H and } 3) = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$$

Guided Practice (20 minutes):

Problem 1:

You flip two coins. List the sample space. What's the probability of getting two tails?

Reasoning:

- Sample space: HH, HT, TH, TT (4 outcomes)
- Favorable: TT (1 outcome)
- $P(\text{TT}) = \frac{1}{4} = 0.25 = 25\%$

Problem 2:

A restaurant offers 3 main dishes (chicken, beef, fish) and 2 sides (rice, salad). Draw a tree diagram. How many meal combinations?

Reasoning:

Main	Side	Combo
Chicken ---	Rice ---	C-

Continue

R

--- Salad --- C-S

Beef ----- Rice --- B-R

--- Salad --- B-S

Fish ----- Rice --- F-R

--- Salad --- F-S

Total: $3 \times 2 = 6$ combinations****Problem 3:****

You spin a spinner (4 equal sections: A, B, C, D) twice. What's the probability of getting A both times?

Reasoning:

$$- P(\text{A}) = \frac{1}{4}$$

$$- P(\text{A and A}) = \frac{1}{4} \times \frac{1}{4} = \frac{1}{16} = 0.0625 = 6.25\%$$

****Quick Check (5 minutes):****

1. Rolling two dice. How many total outcomes?

Answer: 36

2. Flipping 3 coins. What's $P(\text{HHH})$?

*Answer: $\frac{1}{8}$ *

3. If $P(\text{A}) = \frac{1}{3}$ and $P(\text{B}) = \frac{1}{2}$, what's $P(\text{A and B})$?

*Answer: $\frac{1}{6}$ *

****Extension Idea:****

Create a probability game with cards or dice. Calculate all probabilities. Play and track results!

4. Practice Sets (Tiered Difficulty)

Each topic includes three levels. ****Start with the level suggested by the diagnostic.****

TOPIC A: Ratios & Proportional Relationships

Foundation Set (10 items)

1. Find the unit rate: 120 miles in 3 hours
2. Find the unit rate: \$15 for 5 pounds
3. What is 10% of 80?
4. What is 25% of 200?
5. A shirt costs \$40. It's 20% off. What's the discount amount?
6. You tip 15% on a \$20 bill. How much is the tip?
7. Is this proportional? (Check if ratios are equal)

\$x\$	\$y\$	
-----	-----	
2	6	
4	12	

8. A recipe uses 2 cups of flour for 3 servings. How many cups for 1 serving?
9. On a map, 1 inch = 5 miles. The distance on the map is 3 inches. What's the actual distance?
10. If $y = 4x$, what is y when $x = 5$?

Foundation Set-Solutions

1. ****Answer: 40 miles per hour****
Hint: Divide miles by hours
Solution: $120 \div 3 = 40$ mph

2. ****Answer: \$3 per pound****

Solution: $\$15 \div 5 = 3\$$ dollars per pound

3. ****Answer: 8****

Hint: 10% means multiply by 0.10

Solution: $\$80 \times 0.10 = 8\$$

4. ****Answer: 50****

Solution: $\$200 \times 0.25 = 50\$$

5. ****Answer: \$8****

Hint: Find 20% of \$40

Solution: $\$40 \times 0.20 = 8\$$

6. ****Answer: \$3****

Solution: $\$20 \times 0.15 = 3\$$

7. ****Answer: Yes, proportional****

Hint: Check if $\frac{y}{x}$ is constant

Solution: $\frac{6}{2} = 3$, $\frac{12}{4} = 3 \rightarrow$ Constant ratio ✓

8. ****Answer: $\frac{2}{3}$ cup per serving****

Solution: $\$2 \div 3 = \frac{2}{3}$ cup/serving

9. ****Answer: 15 miles****

Hint: Multiply map distance by scale

Solution: $\$3 \times 5 = 15\$$ miles

10. ****Answer: $y = 20$ ****

Solution: $y = 4(5) = 20$

On-Level Set (10 items)

1. Find the unit rate: $\frac{3}{4}$ mile in $\frac{1}{6}$ hour

2. A store sells 3 shirts for \$45. What's the unit price?

3. A \$75 jacket is on sale for 30% off. What's the sale price?

4. You earn \$52 after a 30% raise. What was your original pay?

5. Complete the table for a proportional relationship where $k = 6$:

\$x\$	\$y\$	
-----	-----	
2	?	
?	24	
5	?	

6. A car travels at 65 mph for 3.5 hours. How far does it go?
7. On a scale drawing, 2 cm = 7 m. A room is 5 cm long on the drawing. Actual length?
8. If 8 workers complete a job in 5 days, how long for 1 worker? (Assume constant rate)
9. Write the equation for: "y is proportional to x, and y = 21 when x = 7"
10. Sales tax is 8.5%. You buy an item for \$60. What's the total cost?

On-Level Set-Solutions

1. ****Answer: 4.5 miles per hour****
 Hint: Divide the fractions
 Solution: $\frac{3}{4} \div \frac{1}{6} = \frac{3}{4} \times 6 = \frac{18}{4} = 4.5$ mph
2. ****Answer: \$15 per shirt****
 Solution: $45 \div 3 = 15$ dollars/shirt
3. ****Answer: \$52.50****
 Hint: Find 30% discount, then subtract OR multiply by 0.70
 Solution: $75 \times 0.70 = 52.50$
4. ****Answer: \$40****
 Hint: If \$52 is 130% (100% + 30%), find 100%
 Solution: $52 \div 1.30 = 40$
5. ****Answer:****

\$x\$	\$y\$	
-----	-----	
2	12	
4	24	
5	30	

*Hint: $y = 6x$ *

Solution: For $x=2$: $y = 12$. For $y=24$: $x = 4$. For $x=5$: $y = 30$

6. **Answer: 227.5 miles**

Hint: Distance = rate $\times$ time

Solution: $65 \times 3.5 = 227.5$ miles

7. **Answer: 17.5 m**

*Hint: Set up proportion: $\frac{2}{7} = \frac{5}{x}$ *

Solution: $x = \frac{5 \times 7}{2} = 17.5$ m

8. **Answer: 40 days**

*Hint: Total worker-days = $8 \times 5 = 40$ *

Solution: 1 worker needs 40 days

9. **Answer: $y = 3x$ **

*Hint: Find $k = \frac{y}{x}$ *

Solution: $k = \frac{21}{7} = 3$, so $y = 3x$

10. **Answer: \$65.10**

Hint: Multiply by 1.085

Solution: $60 \times 1.085 = 65.10$

Challenge Set (8 items)

1. A recipe calls for a ratio of flour to sugar of 5:2. If you use 3.5 cups of sugar, how much flour?

2. A population of 12,000 increases by 15% one year, then decreases by 10% the next. Final population?

3. Two quantities are in a 3:4 ratio. If the first quantity is 27, what's the second?

4. A map scale is 1:50,000. Two cities are 12 cm apart on the map. How many kilometers apart in reality? (100,000 cm = 1 km)

5. Store A: \$120 with 25% off. Store B: \$110 with 15% off. Which is cheaper?

6. Complete: If y is inversely proportional to x , and $y = 12$ when $x = 5$, find y when $x = 10$. (Inverse: $xy = k$)

7. A car depreciates 20% per year. It costs \$25,000 new. Value after 2 years?

8. Three partners split profit in ratio 2:3:5. Total profit is \$40,000. How much does each get?

Challenge Set–Solutions

1. ****Answer: 8.75 cups flour****

*Hint: Set up ratio $\frac{5}{2} = \frac{x}{3.5}$ *

Solution: $x = \frac{5 \times 3.5}{2} = \frac{17.5}{2} = 8.75$ cups

2. ****Answer: 12,420****

Common error: Don't add/subtract percents directly!

Solution:

– After 15% increase: $\$12,000 \times 1.15 = \$13,800$

– After 10% decrease: $\$13,800 \times 0.90 = \$12,420$

3. ****Answer: 36****

*Hint: $\frac{3}{4} = \frac{27}{x}$ *

Solution: $x = \frac{27 \times 4}{3} = 36$

4. ****Answer: 6 km****

Hint: Map distance $\times$ scale, then convert cm to km

Solution: $12 \text{ cm} \times 50,000 = 600,000 \text{ cm} = 6 \text{ km}$

5. ****Answer: Store A is cheaper (\$90 vs. \$93.50)****

Solution:

– Store A: $\$120 \times 0.75 = \90

– Store B: $\$110 \times 0.85 = \93.50

6. ****Answer: $y = 6$ ****

Hint: Inverse means $xy = k$ is constant

Solution: $k = 12 \times 5 = 60$. When $x = 10$: $10y = 60$, so $y = 6$

7. ****Answer: \$16,000****

Hint: Apply 80% (after 20% loss) twice

Solution: $\$25,000 \times 0.80 \times 0.80 = \$16,000$

8. ****Answer: \$8,000, \$12,000, \$20,000****

*Hint: Total parts = $2+3+5 = 10$ *

Solution:

– First: $\frac{2}{10} \times 40,000 = \$8,000$

– Second: $\frac{3}{10} \times 40,000 = \$12,000$

– Third: $\frac{5}{10} \times 40,000 = \$20,000$

TOPIC B: Operations with Rational Numbers

Foundation Set (10 items)

1. $-5 + 8 =$ \$
2. $7 - 12 =$ \$
3. $-3 + (-6) =$ \$
4. $-4 - (-9) =$ \$
5. $(-6) \times 3 =$ \$
6. $(-20) \div (-4) =$ \$
7. $\frac{1}{2} + \frac{1}{4} =$ \$
8. $\frac{3}{4} - \frac{1}{6} =$ \$
9. $\frac{2}{3} \times \frac{3}{5} =$ \$
10. $\frac{1}{2} \div \frac{1}{4} =$ \$

Foundation Set-Solutions

1. ****Answer: 3****
 Hint: Different signs, subtract absolute values
 Solution: $8 - 5 = 3$ (positive because $8 > 5$)
2. ****Answer: -5****
 Solution: $7 + (-12) = -5$
3. ****Answer: -9****
 Hint: Same signs, add and keep sign
 Solution: $3 + 6 = 9$, keep negative $\rightarrow -9$
4. ****Answer: 5****
 Hint: Subtracting a negative = adding a positive
 Solution: $-4 + 9 = 5$
5. ****Answer: -18****
 Hint: Different signs $\rightarrow$ negative

Solution: $\$6 \times 3 = 18\$$, make negative $\rightarrow \$-18\$$

6. ****Answer: 5****

Hint: Same signs $\rightarrow$ positive

Solution: $\$20 \div 4 = 5\$$

7. ****Answer: $\frac{3}{4}$ ****

Hint: LCD = 4

Solution: $\frac{2}{4} + \frac{1}{4} = \frac{3}{4}$

8. ****Answer: $\frac{7}{12}$ ****

Hint: LCD = 12

Solution: $\frac{9}{12} - \frac{2}{12} = \frac{7}{12}$

9. ****Answer: $\frac{2}{5}$ ****

Solution: $\frac{2 \times 3}{3 \times 5} = \frac{6}{15} = \frac{2}{5}$

10. ****Answer: 2****

Hint: Keep, change, flip

Solution: $\frac{1}{2} \times \frac{4}{1} = 2$

On-Level Set (10 items)

1. $\$-15 + 7 - 3 = \$$

2. $\$-8 - (-12) + 5 = \$$

3. $\$(-3) \times (-7) \times 2 = \$$

4. $\frac{-24}{6} \times (-2) = \$$

5. $\frac{5}{6} + \frac{2}{3} - \frac{1}{2} = \$$

6. $2\frac{1}{4} - 1\frac{3}{4} = \$$

7. $\$(-\frac{2}{3}) \times \frac{9}{4} = \$$

8. $\frac{5}{8} \div (-\frac{1}{4}) = \$$

9. $\$0.75 + (-1.2) = \$$

10. $\$-3.6 \div 0.9 = \$$

On-Level Set-Solutions

1. **Answer: -11**

Common error: Forgetting to work left to right

Solution: $\$-15 + 7 = -8$, then $\$-8 - 3 = -11$

2. **Answer: 9**

Solution: $\$-8 + 12 + 5 = 9$

3. **Answer: 42**

Hint: Two negatives make positive, then multiply by 2

Solution: $\$(-3) \times (-7) = 21$, then $\$21 \times 2 = 42$

4. **Answer: 8**

Solution: $\$\frac{-24}{6} = -4$, then $\$(-4) \times (-2) = 8$

5. **Answer: 1**

Hint: LCD = 6

Solution: $\$\frac{5}{6} + \frac{4}{6} - \frac{3}{6} = \frac{6}{6} = 1$

6. **Answer: $\frac{1}{2}$ **

Solution: $\$\frac{9}{4} - \frac{7}{4} = \frac{2}{4} = \frac{1}{2}$

7. **Answer: $-\frac{3}{2}$ or -1.5 **

Solution: $\$\frac{-2 \times 9}{3 \times 4} = \frac{-18}{12} = -\frac{3}{2}$

8. **Answer: $-\frac{5}{2}$ or -2.5 **

Hint: Keep, change, flip

Solution: $\$\frac{5}{8} \times (-4) = -\frac{20}{8} = -\frac{5}{2}$

9. **Answer: -0.45**

Solution: $\$0.75 + (-1.2) = -0.45$

10. **Answer: -4**

Solution: $\$-3.6 \div 0.9 = -4$

Challenge Set (8 items)

1. Evaluate: $\$-5 + 3 \times (-2) - 4 \div 2$

2. $\$(-\frac{3}{4}) \div (-\frac{2}{5}) \times \frac{5}{6} = \$$

3. $\$2.5 \times (-1.6) + 3.2 \div (-0.8) = \$$

4. Simplify: $\frac{-12 + 4 \times 3}{-2 \times 5}$
5. $3\frac{1}{3} \times (-1\frac{1}{5}) =$
6. The temperature was -8°F . It rose 15° , then dropped 20° . Final temperature?
7. A submarine at -120 m dives 45 m deeper, then rises 80 m. Final depth?
8. Calculate: $|-7| - |3| + |-5 + 2|$

Challenge Set-Solutions

1. ****Answer: -13****
 Hint: Order of operations! Multiply and divide first
 Solution: $-5 + (3 \times -2) - (4 \div 2) = -5 + (-6) - 2 = -13$
2. ****Answer: $\frac{25}{16}$ or $1\frac{9}{16}$ ****
 Solution:
 - $(-\frac{3}{4}) \div (-\frac{2}{5}) = \frac{3}{4} \times \frac{5}{2} = \frac{15}{8}$
 - $\frac{15}{8} \times \frac{5}{6} = \frac{75}{48} = \frac{25}{16}$
3. ****Answer: -8****
 Solution: $(2.5 \times -1.6) + (3.2 \div -0.8) = -4 + (-4) = -8$
4. ****Answer: 0****
 Solution:
 - Numerator: $-12 + 12 = 0$
 - Denominator: -10
 - $\frac{0}{-10} = 0$
5. ****Answer: -4****
 Solution: $\frac{10}{3} \times (-\frac{6}{5}) = \frac{-60}{15} = -4$
6. ****Answer: -13°F ****
 Solution: $-8 + 15 = 7$, then $7 - 20 = -13$
7. ****Answer: -85 m****
 Common error: Confusing "deeper" (negative) with positive
 Solution: $-120 + (-45) + 80 = -85$ m
8. ****Answer: 7****
 Hint: Absolute value makes everything positive

Solution: $7 - 3 + |-3| = 7 - 3 + 3 = 7$

TOPIC C: Expressions, Equations & Inequalities

Foundation Set (10 items)

1. Combine like terms: $5x + 3x$
2. Simplify: $7y - 2y + 4y$
3. Expand: $3(x + 5)$
4. Expand: $-2(4 - y)$
5. Solve: $x + 7 = 15$
6. Solve: $3x = 21$
7. Solve: $2x + 5 = 13$
8. Solve: $x - 4 = -10$
9. Solve: $\frac{x}{3} = 6$
10. Solve: $4x - 3 = 9$

Foundation Set-Solutions

1. **Answer:** $8x$
Solution: $5 + 3 = 8$
2. **Answer:** $9y$
Solution: $7 - 2 + 4 = 9$
3. **Answer:** $3x + 15$
Solution: $3 \times x + 3 \times 5$
4. **Answer:** $-8 + 2y$
Hint: Distribute the negative
Solution: $-2 \times 4 = -8$, $-2 \times (-y) = 2y$
5. **Answer:** $x = 8$

Solution: $\$15 - 7 = 8\$$

6. ****Answer:** $\$x = 7\$$

Solution: $\$21 \div 3 = 7\$$

7. ****Answer:** $\$x = 4\$$

Solution: $\$2x = 8\$$, then $\$x = 4\$$

8. ****Answer:** $\$x = -6\$$

Solution: $\$-10 + 4 = -6\$$

9. ****Answer:** $\$x = 18\$$

Solution: $\$6 \times 3 = 18\$$

10. ****Answer:** $\$x = 3\$$

Solution: $\$4x = 12\$$, then $\$x = 3\$$

On-Level Set (10 items)

1. Simplify: $\$4(2x - 3) + 5x\$$

2. Simplify: $\$-3(x + 4) - 2x + 8\$$

3. Solve: $\$5x - 8 = 22\$$

4. Solve: $\$\frac{x}{5} + 3 = 7\$$

5. Solve: $\$-3x + 10 = 1\$$

6. Solve: $\$7 - 2x = 15\$$

7. Solve the inequality: $\$x + 6 < 10\$$

8. Solve the inequality: $\$3x \geq 15\$$

9. Write an equation: "Five more than twice a number is 17."

10. Write an equation: "Three times a number, decreased by 7, equals 20."

On-Level Set-Solutions

1. ****Answer:** $\$13x - 12\$$

Solution: $8x - 12 + 5x = 13x - 12$

2. **Answer:** $-5x - 4$

Hint: Distribute, then combine

Solution: $-3x - 12 - 2x + 8 = -5x - 4$

3. **Answer:** $x = 6$

Solution: $5x = 30$, $x = 6$

4. **Answer:** $x = 20$

Solution: $\frac{x}{5} = 4$, $x = 20$

5. **Answer:** $x = 3$

Solution: $-3x = -9$, $x = 3$

6. **Answer:** $x = -4$

Solution: $-2x = 8$, $x = -4$

7. **Answer:** $x < 4$

Solution: $x < 10 - 6$

8. **Answer:** $x \geq 5$

Solution: $x \geq 15 \div 3$

9. **Answer:** $2x + 5 = 17$

Hint: "Twice a number" = $2x$, "five more" = $+5$

10. **Answer:** $3x - 7 = 20$

Challenge Set (8 items)

1. Simplify: $5(2x - 3) - 3(x + 4) + 7x$

2. Solve: $3(x + 2) - 5 = 16$

3. Solve: $\frac{2x - 5}{3} = 7$

4. Solve the inequality: $-4x + 7 < -5$

5. Solve: $2(3x - 4) = 5x + 10$

6. Write and solve: "The sum of a number and its double is 21. Find the number."

7. Solve: $\frac{x}{4} - \frac{x}{6} = 1$

8. Solve the inequality and graph: $5 - 2x \leq 11$

Challenge Set–Solutions

1. **Answer:** $14x - 27$

Solution: $10x - 15 - 3x - 12 + 7x = 14x - 27$

2. **Answer:** $x = 5$

Solution: $3x + 6 - 5 = 16$, $3x + 1 = 16$, $3x = 15$, $x = 5$

3. **Answer:** $x = 13$

Solution: $2x - 5 = 21$, $2x = 26$, $x = 13$

4. **Answer:** $x > 3$

Hint: When dividing by negative, flip the sign!

Solution: $-4x < -12$, $x > 3$

5. **Answer:** $x = 18$

Solution: $6x - 8 = 5x + 10$, $x = 18$

6. **Answer:** 7

*Hint: Let x = the number. Equation: $x + 2x = 21$ *

Solution: $3x = 21$, $x = 7$

7. **Answer:** $x = 12$

Hint: LCD = 12

Solution: $\frac{3x}{12} - \frac{2x}{12} = 1$, $\frac{x}{12} = 1$, $x = 12$

8. **Answer:** $x \geq -3$

Solution: $-2x \leq 6$, $x \geq -3$ (flip sign when dividing by negative)

Graph: closed circle at -3 , shade to the right $\rightarrow$

TOPIC D: Geometry

Foundation Set (10 items)

1. Find circumference: radius = 4 cm ($\pi \approx 3.14$)

2. Find area of circle: radius = 3 m ($\pi \approx 3.14$)

3. Find volume of rectangular prism: $5 \times 3 \times 2$ cm

4. Find volume of cube: edge = 4 in
5. A triangle has base 6 cm, height 5 cm. Area?
6. Find surface area of cube: edge = 3 ft
7. Complementary angles add to 90° . If one is 35° , find the other.
8. Find volume of cylinder: radius = 2 cm, height = 10 cm ($\pi \approx 3.14$)
9. Scale: 1 cm = 3 m. Drawing length = 4 cm. Actual length?
10. Surface area of rectangular prism: $2 \times 3 \times 5$ cm

Foundation Set–Solutions

1. **Answer:** 25.12 cm
Hint: $C = 2\pi r$
Solution: $2 \times 3.14 \times 4 = 25.12$ cm
2. **Answer:** 28.26 m²
Hint: $A = \pi r^2$
Solution: $3.14 \times 3^2 = 28.26$ m²
3. **Answer:** 30 cm³
Solution: $5 \times 3 \times 2 = 30$ cm³
4. **Answer:** 64 in³
Solution: $4^3 = 64$ in³
5. **Answer:** 15 cm²
Hint: $A = \frac{1}{2}bh$
Solution: $\frac{1}{2} \times 6 \times 5 = 15$ cm²
6. **Answer:** 54 ft²
Hint: 6 faces, each 3×3
Solution: $6 \times 9 = 54$ ft²
7. **Answer:** 55°
Solution: $90 - 35 = 55^\circ$
8. **Answer:** 125.6 cm³
Hint: $V = \pi r^2 h$

Solution: $3.14 \times 2^2 \times 10 = 125.6 \text{ cm}^3$

9. **Answer: 12 m**

Solution: $4 \times 3 = 12 \text{ m}$

10. **Answer: 62 cm²**

*Hint: $SA = 2(lw + lh + wh)$

Solution: $2(6 + 10 + 15) = 2(31) = 62 \text{ cm}^2$

On-Level Set (10 items)

- Find area of circle: diameter = 10 inches ($\pi \approx 3.14$)
- A cylinder has diameter 6 cm, height 8 cm. Find volume. ($\pi \approx 3.14$)
- Surface area of rectangular prism: $4 \times 5 \times 6 \text{ m}$
- Find volume of triangular prism: base area = 12 cm^2 , height = 10 cm
- Two angles are supplementary. One is 115° . Find the other.
- Vertical angles are equal. If one is 72° , what's the other?
- Scale 1:200. Model car is 8 cm long. Actual car length?
- A circular pool has circumference 31.4 m. Find the radius. ($\pi \approx 3.14$)
- Square pyramid: base side = 6 in, triangular face height = 8 in. Surface area?
- Find volume of pyramid: base area = 20 cm^2 , height = 9 cm

On-Level Set-Solutions

1. **Answer: 78.5 in²**

Hint: Use radius = 5 in

Solution: $3.14 \times 5^2 = 78.5 \text{ in}^2$

2. **Answer: 226.08 cm³**

Hint: Radius = 3 cm

Solution: $3.14 \times 3^2 \times 8 = 226.08 \text{ cm}^3$

3. **Answer: 148 m²**

Solution: $2(20 + 24 + 30) = 148 \text{ m}^2$

4. **Answer:** 120 cm^3

Hint: $V = Bh$ where B = base area

Solution: $12 \times 10 = 120 \text{ cm}^3$

5. **Answer:** 65°

*Hint: Supplementary = 180° *

Continue

Solution: $180 - 115 = 65^\circ$

6. **Answer:** 72°

Solution: Vertical angles are congruent

7. **Answer:** 1,600 cm or 16 m *Hint: Scale 1:200 means multiply by 200* Solution: $8 \times 200 = 1,600 \text{ cm}$

8. **Answer:** 5 m *Hint: $C = 2\pi r$, solve for r * Solution: $31.4 = 2 \times 3.14 \times r, r = 5 \text{ m}$

9. **Answer:** 132 in^2

Solution:

- Base: $6^2 = 36 \text{ in}^2$
- 4 triangles: $4 \times (\frac{1}{2} \times 6 \times 8) = 96 \text{ in}^2$
- Total: $36 + 96 = 132 \text{ in}^2$

10. **Answer:** 60 cm^3 *Hint: $V = \frac{1}{3}Bh$ * Solution: $\frac{1}{3} \times 20 \times 9 = 60 \text{ cm}^3$

Challenge Set (8 items)

1. A circle has area 78.5 cm^2 . Find its circumference. ($\pi \approx 3.14$)
2. Cone volume: radius = 3 in, height = 7 in. ($V = \frac{1}{3}\pi r^2 h, \pi \approx 3.14$)
3. A cylinder and cone have the same radius and height. The cylinder volume is 150 cm^3 . What's the cone volume?
4. Scale drawing: 1 inch = 4 feet. A room is $3.5 \text{ in} \times 2.5 \text{ in}$ on the drawing. Actual area?
5. Surface area of cylinder: radius = 5 cm, height = 12 cm. ($SA = 2\pi r^2 + 2\pi r h, \pi \approx 3.14$)

6. Three angles in a triangle measure x , $2x$, and $3x$. Find each angle.
 7. A rectangular prism has volume 120 cm^3 . Length = 6 cm, width = 4 cm. Find height.
 8. A circle is inscribed in a square with side 10 cm. What's the circle's area? ($\pi \approx 3.14$)
-

Challenge Set—Solutions

1. ****Answer: 31.4 cm**** *Hint: Find radius from area first* Solution: $\pi r^2 = 78.5$, $r^2 = 25$, $r = 5$.
Then $C = 2\pi r = 31.4 \text{ cm}$
 2. **Answer: 65.94 in³**
Solution: $V = \frac{1}{3} \times 3.14 \times 3^2 \times 7 = 65.94 \text{ in}^3$
 3. ****Answer: 50 cm³**** *Hint: Cone volume = $\frac{1}{3} \times$ cylinder volume* Solution: $150 \div 3 = 50 \text{ cm}^3$
 4. **Answer: 140 ft²**
Common error: Forgetting to square the scale for area
Solution:
 - Actual dimensions: $3.5 \times 4 = 14 \text{ ft}$, $2.5 \times 4 = 10 \text{ ft}$
 - Area: $14 \times 10 = 140 \text{ ft}^2$
 5. **Answer: 533.8 cm²**
Solution: $SA = 2(3.14 \times 25) + 2(3.14 \times 5 \times 12) = 157 + 376.8 = 533.8 \text{ cm}^2$
 6. ****Answer: 30°, 60°, 90°**** *Hint: Triangle angles sum to 180°* Solution: $x + 2x + 3x = 180$,
 $6x = 180$, $x = 30$
 7. **Answer: 5 cm**
Solution: $6 \times 4 \times h = 120$, $24h = 120$, $h = 5 \text{ cm}$
 8. ****Answer: 78.5 cm²**** *Hint: Circle diameter = square side* Solution: Radius = 5 cm, $A =$
 $3.14 \times 5^2 = 78.5 \text{ cm}^2$
-

TOPIC E: Statistics & Probability

Foundation Set (10 items)

1. Data: 5, 7, 8, 10, 10. Find the mean.
2. Data: 3, 6, 7, 9, 10. Find the median.

3. Data: 2, 2, 5, 2, 8. Find the mode.
 4. Data: 15, 20, 18, 22. Find the range.
 5. Flip a fair coin. $P(\text{heads}) = ?$
 6. Roll a die. $P(5) = ?$
 7. A bag has 4 red, 3 blue, 2 green marbles. $P(\text{red}) = ?$
 8. Flip a coin twice. How many outcomes?
 9. Roll a die. $P(\text{even number}) = ?$
 10. A spinner has 8 equal sections: 3 red, 5 blue. $P(\text{blue}) = ?$
-

Foundation Set—Solutions

1. Answer: 8

Solution: $(5 + 7 + 8 + 10 + 10) \div 5 = 40 \div 5 = 8$

2. Answer: 7

Hint: Middle value

Solution: Already ordered, median = 7

3. Answer: 2

Hint: Most frequent

Solution: 2 appears 3 times

4. Answer: 7

Solution: $22 - 15 = 7$

5. Answer: $\frac{1}{2}$ or 0.5

6. Answer: $\frac{1}{6}$

7. Answer: $\frac{4}{9}$

Solution: Total = 9 marbles

8. Answer: 4

Hint: HH, HT, TH, TT

9. **Answer: $\frac{1}{2}$ or $\frac{3}{6}$ **Hint: Even = 2, 4, 6*

10. Answer: $\frac{5}{8}$

On-Level Set (10 items)

1. Data: 12, 15, 18, 20, 35. Mean or median better? Why?
 2. Data: 4, 7, 9, 11, 14. Find mean and median.
 3. You flip a coin 40 times, get 18 heads. Experimental P (heads)?
 4. Two dice rolled. What's P (sum = 7)? (Use sample space or reasoning)
 5. A bag has 5 red, 3 blue marbles. Draw one, don't replace, draw again. P (2 reds)?
 6. Data: 10, 12, 15, 15, 18, 20. Find interquartile range (IQR). (IQR = $Q_3 - Q_1$)
 7. Spinner: 6 equal sections (2 yellow, 4 green). Spin twice. P (both green)?
 8. Which data set has more variability? A: 5, 6, 7, 8, 9 or B: 2, 5, 10, 15, 18?
 9. Create a sample space: Flip coin, roll die. How many outcomes?
 10. True or False: If P (event) = 0.8, the event is certain.
-

On-Level Set—Solutions**1. Answer: Median is better (35 is an outlier)**

Explanation: Mean = 20, Median = 18. The outlier 35 pulls mean up.

2. Answer: Mean = 9, Median = 9

Solution: $(4 + 7 + 9 + 11 + 14) \div 5 = 45 \div 5 = 9$

3. Answer: $\frac{18}{40} = \frac{9}{20} = 0.45$

4. **Answer: $\frac{1}{6}$ ** *Hint: 6 ways to get 7: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) out of 36 total* Solution: $\frac{6}{36} = \frac{1}{6}$

5. **Answer: $\frac{5}{14}$ ** *Hint: First draw: $\frac{5}{8}$, second draw: $\frac{4}{7}$ * Solution: $\frac{5}{8} \times \frac{4}{7} = \frac{20}{56} = \frac{5}{14}$

6. Answer: 5

Hint: $Q_1 = 12$, $Q_3 = 17$ (average of 15 and 18 if needed, but here use 18 and 15 positions)

Solution: Lower half: 10, 12, 15 $\rightarrow Q_1 = 12$. Upper half: 15, 18, 20 $\rightarrow Q_3 = 18$. IQR = $18 - 12 = 6$

(Note: Q_1/Q_3 calculation can vary by method; accept 5 or 6)

7. Answer: $\frac{4}{9}$

Solution: P (green) = $\frac{4}{6} = \frac{2}{3}$, P (both) = $\frac{2}{3} \times \frac{2}{3} = \frac{4}{9}$

8. Answer: B

Hint: Larger range = more variability

Solution: A has range 4, B has range 16

9. Answer: 12 outcomes

Solution: $2 \text{ (coin)} \times 6 \text{ (die)} = 12$

10. Answer: False

Hint: Certain = probability of 1

Explanation: 0.8 is likely, but not certain

Challenge Set (8 items)

1. Data: 20, 22, 25, 28, 100. Calculate mean with and without the outlier. Compare.
2. Two data sets have the same mean (10). Set A range = 4, Set B range = 20. Which is more consistent?
3. You draw 2 cards from a deck without replacement. P (both aces)?
4. A test has 5 true/false questions. If you guess randomly, what's the probability of getting all 5 correct?
5. Given: $P(\text{rain}) = 0.4$ each day. Assuming independence, $P(\text{rain both Saturday and Sunday})$?
6. Data: 5, 8, 10, 12, 15, 18, 20. Find the mean absolute deviation (MAD).
 $MAD = \text{average of } |\text{each value} - \text{mean}|$
7. Three coins flipped. $P(\text{exactly 2 heads})$?
8. In a group of 30 students, 18 like math, 20 like science, 12 like both. If you pick a student at random, $P(\text{likes math or science})$?

Challenge Set—Solutions**1. Answer: With outlier: 39; Without: 23.75**

Solution:

- With: $(20 + 22 + 25 + 28 + 100) \div 5 = 195 \div 5 = 39$

- Without: $(20 + 22 + 25 + 28) \div 4 = 95 \div 4 = 23.75$

The outlier significantly raises the mean.

2. Answer: Set A

Explanation: Smaller range = less variability = more consistent

3. Answer: $\frac{1}{221}$

Solution: $P(\text{first ace}) = \frac{4}{52}$, $P(\text{second ace}) = \frac{3}{51}$

$$\frac{4}{52} \times \frac{3}{51} = \frac{12}{2652} = \frac{1}{221}$$

4. **Answer: $\frac{1}{32}$ ** *Hint: Each question has $\frac{1}{2}$ probability* Solution: $(\frac{1}{2})^5 = \frac{1}{32}$

5. Answer: 0.16

Solution: $0.4 \times 0.4 = 0.16$

6. Answer: 4

Solution:

- Mean = 12
 - Deviations: $|5-12|=7$, $|8-12|=4$, $|10-12|=2$, $|12-12|=0$, $|15-12|=3$, $|18-12|=6$, $|20-12|=8$
 - Sum = $7+4+2+0+3+6+8 = 30$
 - MAD = $30 \div 7 \approx 4.29$ (accept 4 or 4.3)
7. **Answer: $\frac{3}{8}$ ** *Hint: Sample space = 8 outcomes, 3 have exactly 2 heads: HHT, HTH, THH* Solution: $\frac{3}{8}$
8. **Answer: $\frac{26}{30} = \frac{13}{15}$ ** *Hint: Use inclusion-exclusion: $P(M \text{ or } S) = P(M) + P(S) - P(M \text{ and } S)$ * Solution: $18 + 20 - 12 = 26$ students. $\frac{26}{30}$

5. Real-World Tasks & Projects

Each project includes materials, instructions, and a grading checklist.

Project A1: Design Your Dream Bedroom (Scale Drawing)

Topic: Geometry (Scale Drawings, Area, Perimeter)

Materials:

- Graph paper (or plain paper and ruler)
- Pencil, eraser
- Ruler or measuring tape
- Calculator

Instructions:

1. **Measure:** Measure an actual room in your house (or use imaginary dimensions 12 ft × 10 ft).
2. **Choose a scale:** Use 1 cm = 1 ft (or 1 inch = 2 ft).
3. **Draw the floor plan:** Draw the room outline to scale on graph paper. Label dimensions.
4. **Add furniture:** Draw at least 5 pieces of furniture to scale (bed, desk, dresser, etc.). You can look up typical furniture sizes online.
5. **Calculate:**
 - Actual room area (in square feet)
 - Actual room perimeter
 - Scale drawing area (in square cm or inches)
6. **Reflection:** Write 2-3 sentences: Did everything fit? What did you learn about scale?

Grading Checklist:

- ☐ Scale is clearly stated and used correctly (2 pts)
- ☐ Room dimensions are labeled (1 pt)
- ☐ At least 5 furniture pieces drawn to scale (2 pts)
- ☐ Area and perimeter calculated correctly (2 pts)
- ☐ Drawing is neat and clear (1 pt)
- ☐ Reflection included (2 pts)

Total: 10 pts

Project A2: Budget Shopping with Discounts**Topic:** Ratios & Proportional Relationships (Percent, Discounts, Tax)**Materials:**

- Store ads or online shopping sites

- Calculator
- Paper for recording

Instructions:

1. **Choose 5 items** you'd like to buy (clothes, games, etc.). Find the original price for each.
2. **Apply discounts:** Each item is on sale. Find or imagine discounts (e.g., 20% off, 15% off, 30% off).
3. **Calculate sale prices** for each item.
4. **Add sales tax:** Use your local sales tax rate (or use 8%).
5. **Find the total cost** after discounts and tax.
6. **Bonus:** How much did you save compared to buying everything at full price with tax?

Grading Checklist:

- ☐ 5 items listed with original prices (1 pt)
- ☐ Discounts correctly applied for each item (3 pts)
- ☐ Sale prices calculated correctly (2 pts)
- ☐ Tax added correctly (2 pts)
- ☐ Total cost calculated (1 pt)
- ☐ Bonus: Savings calculated (1 pt)

Total: 10 pts

Project B1: Integer Temperature Tracker**Topic:** Operations with Rational Numbers (Integers)**Materials:**

- Internet access (weather website)
- Graph paper or spreadsheet
- Calculator

Instructions:

1. **Track temperatures:** Record the daily high and low temperatures for 7 days. Use a location with temperature variation (or use historical data from a cold month).
2. **Calculate daily range:** For each day, find the difference between high and low.
3. **Calculate weekly mean:** Find the mean high temperature and mean low temperature.
4. **Graph it:** Create a line graph showing high and low temperatures across the 7 days.
5. **Integer operations:** Calculate:
 - The change in high temperature from Day 1 to Day 7
 - The coldest temperature overall
 - If temperature drops 15° from the warmest day, what would it be?

Grading Checklist:

- ☐ 7 days of data collected (high and low) (2 pts)
- ☐ Daily ranges calculated (1 pt)
- ☐ Mean high and low calculated (2 pts)
- ☐ Graph is clear and labeled (2 pts)
- ☐ Integer operations completed correctly (2 pts)
- ☐ Presentation is organized (1 pt)

Total: 10 pts

Project C1: Solving Real-Life Equations

Topic: Expressions, Equations & Inequalities

Materials:

- Paper, pencil
- Calculator

Instructions:

1. **Write 3 word problems** from your life that involve two-step equations. Examples:
 - "I earned money babysitting. After buying a \$15 book, I had \$32 left. How much did I earn?"
 - "A taxi charges \$3 plus \$2 per mile. My ride cost \$19. How many miles?"

- "I'm thinking of a number. If I triple it and subtract 7, I get 20. What's the number?"

2. For each problem:

- Define the variable
- Write the equation
- Solve it step-by-step
- Check your answer
- Answer the question in a complete sentence

3. **Challenge:** Create one problem that needs an inequality (e.g., "I have \$50 to spend...").

Grading Checklist:

- ☐ 3 original word problems created (3 pts)
- ☐ Variables defined for each (1 pt)
- ☐ Equations written correctly (2 pts)
- ☐ Solutions shown with steps (2 pts)
- ☐ Answers checked and stated in sentences (1 pt)
- ☐ Challenge inequality problem included (1 pt)

Total: 10 pts

Project D1: Volume Investigation—Which Container Holds More?

Topic: Geometry (Volume)

Materials:

- 3 different containers (cylinder, rectangular prism, or irregular shapes—cans, boxes, jars)
- Ruler or measuring tape
- Calculator
- Water and measuring cup (for verification)

Instructions:

1. **Measure each container:** Record dimensions (length, width, height, radius as needed).
2. **Calculate volume** for each using formulas.

3. **Rank them** from least to greatest volume based on your calculations.
4. **Verify (optional but fun):** Fill each with water, measure the water. Compare to your calculations.
5. **Reflection:** Were your calculations close to actual? Why might they differ?

Grading Checklist:

- ☐ 3 containers measured with dimensions labeled (2 pts)
- ☐ Correct formulas used for each shape (2 pts)
- ☐ Volumes calculated correctly (3 pts)
- ☐ Containers ranked by volume (1 pt)
- ☐ Verification attempted and results discussed (1 pt)
- ☐ Reflection included (1 pt)

Total: 10 pts

Project E1: Probability Experiment—Coin & Die**Topic:** Statistics & Probability**Materials:**

- 1 coin
- 1 die
- Paper for recording
- Calculator (optional)

Instructions:

1. **Predict:** Before you start, predict the theoretical probability of:
 - Getting heads AND rolling a 6
 - Getting tails AND rolling an even number
2. **Experiment:** Flip the coin and roll the die together 30 times. Record each outcome.
3. **Calculate experimental probability** for the two events above.
4. **Compare:** How close were your experimental results to the theoretical predictions?
5. **Graph:** Create a bar graph showing how many times each coin-die combination occurred.

Grading Checklist:

- ☐ Theoretical probabilities calculated correctly (2 pts)
- ☐ 30 trials conducted and recorded (2 pts)
- ☐ Experimental probabilities calculated (2 pts)
- ☐ Comparison and explanation (2 pts)
- ☐ Bar graph included and labeled (2 pts)

Total: 10 pts

Project E2: Survey & Data Analysis**Topic:** Statistics & Probability**Materials:**

- Survey question (create one)
- At least 15 people to survey
- Paper, calculator
- Graph paper or computer

Instructions:

1. **Create a question:** Something with numerical answers (e.g., "How many hours of screen time per day?" or "How many siblings do you have?").
2. **Collect data:** Survey at least 15 people. Record responses.
3. **Analyze:**
 - Calculate mean, median, mode, range
 - Identify any outliers
 - Decide which measure of center best represents your data
4. **Visualize:** Create a dot plot, histogram, or bar graph.
5. **Write a summary:** 4-5 sentences about what you learned.

Grading Checklist:

- ☐ Survey question is clear (1 pt)

- ☐ At least 15 responses collected (1 pt)
- ☐ Mean, median, mode, range calculated (3 pts)
- ☐ Outliers identified and measure of center justified (2 pts)
- ☐ Graph is clear and labeled (2 pts)
- ☐ Summary included (1 pt)

Total: 10 pts

6. Mastery Checks & Spiral Review

Weekly Mastery Check Template (8–10 items, 20 minutes)

Use after completing a topic or at end of each week to assess understanding.

Week 1 Mastery Check (Topics A & B focus)

1. Find the unit rate: 360 miles in 6 hours
2. A \$50 item is 15% off. What's the sale price?
3. Calculate: $-7 + 12$
4. Calculate: $(-3) \times (-8)$
5. Solve: $4x - 9 = 15$
6. Simplify: $5(2x - 1) + 3x$
7. Find circumference: radius = 7 cm ($\pi \approx 3.14$)
8. Data: 6, 8, 10, 12, 14. Find the mean.

Performance Bands:

- **7–8 correct:** Mastery! Ready to move on.
 - **5–6 correct:** Solid understanding. Review missed concepts briefly.
 - **3–4 correct:** Partial mastery. Re-teach weak areas before moving on.
 - **0–2 correct:** Needs significant review. Revisit lessons and practice sets.
-

4-Week Spiral Review Plan

Purpose: Mix topics to build retention and prevent forgetting.

Week 1: Focus on Topic A (Ratios) + Topic B (Rational Numbers)

- Practice Set: 5 problems from A, 5 from B
- Include one real-world application

Week 2: Add Topic C (Equations)

- Practice Set: 3 from A, 3 from B, 4 from C
- Include one project element

Week 3: Add Topic D (Geometry)

- Practice Set: 2 from A, 2 from B, 3 from C, 3 from D
- Focus on mixed word problems

Week 4: Add Topic E (Statistics) + Full Review

- Practice Set: 2 from each topic (10 total)
- Include cumulative mastery check
- Review any persistent weak areas

Sample Spiral Review Problems (Week 4):

1. (Topic A) If 5 notebooks cost \$12, what's the unit price?
2. (Topic B) Calculate: $\frac{3}{4} \div \frac{2}{5}$
3. (Topic C) Solve: $\frac{x}{3} + 5 = 11$
4. (Topic D) Find volume: rectangular prism $4 \times 5 \times 3$ cm
5. (Topic E) Data: 5, 7, 9, 11. Find median.
6. (Topic A) A shirt is \$40, then 20% off, then 10% off that price. Final cost?
7. (Topic B) $\$-15 + (-8) - 3 = \$$
8. (Topic C) Write an equation: "Six less than twice a number is 14."
9. (Topic D) Circle area: diameter = 12 in ($\pi \approx 3.14$)

10. (Topic E) Roll a die. $P(\text{number} > 4) = ?$

7. Strategy Guides

Guide 1: Fractions & Negatives Fluency

Fractions Quick Tips:

Adding/Subtracting:

1. Find LCD (Least Common Denominator)
2. Convert each fraction
3. Add/subtract numerators
4. Simplify if possible

Example: $\frac{3}{8} + \frac{1}{6}$

LCD = 24

$$\frac{9}{24} + \frac{4}{24} = \frac{13}{24}$$

Multiplying:

- Multiply straight across
- Simplify before or after

Dividing:

- Keep, Change, Flip (KCF)
- Then multiply

Negatives Quick Tips:

Adding:

- Same signs: Add, keep sign
- Different signs: Subtract, take sign of larger

Subtracting:

- Change to "adding the opposite"

- $5 - (-3) = 5 + 3 = 8$

Multiplying/Dividing:

- Same signs → Positive
- Different signs → Negative

Worked Example: $-\frac{2}{3} \div \frac{4}{5}$

- 1. Keep, Change, Flip: $-\frac{2}{3} \times \frac{5}{4}$
- 2. Multiply: $\frac{-10}{12}$
- 3. Simplify: $-\frac{5}{6}$

Guide 2: Translating Word Problems to Equations

Step-by-Step Strategy:

Step 1: Read carefully

- Identify what you're solving for
- Look for key words

Step 2: Define your variable

- Write "Let $x = \dots$ " clearly

Step 3: Translate phrases

Phrase	Math
sum, total, more than, increased by	+
difference, less than, decreased by, fewer	-
product, times, of (with fractions)	×
quotient, divided by, per, out of	÷
is, equals, gives, results in	=

Step 4: Write the equation

- Match the sentence structure to math

Step 5: Solve and check**Worked Examples:****Example 1:** "Four more than three times a number is 19."

- Let x = the number
- "Three times a number" = $3x$
- "Four more than" = $3x + 4$
- "is 19" = $= 19$
- Equation: $3x + 4 = 19$
- Solve: $3x = 15, x = 5$
- Check: $3(5) + 4 = 19 \checkmark$

Example 2: "Jose has \$5 more than twice what Maria has. Together they have \$50."

- Let m = Maria's money
- Jose = $2m + 5$
- Equation: $m + (2m + 5) = 50$
- Solve: $3m + 5 = 50, 3m = 45, m = 15$
- Maria has \$15, Jose has \$35

Guide 3: Proportion & Percent Problem Heuristics**Proportion Problems:****Set up the proportion:**

$$\frac{\text{part 1}}{\text{part 2}} = \frac{\text{corresponding 1}}{\text{corresponding 2}}$$

Cross multiply:

$$ad = bc \text{ if } \frac{a}{b} = \frac{c}{d}$$

Example: If 3 apples cost \$6, how much do 8 apples cost?

$$\frac{3 \text{ apples}}{6 \text{ dollars}} = \frac{8 \text{ apples}}{x \text{ dollars}}$$

Cross multiply: $3x = 48, x = 16$

Percent Problems—Three Types:

Type 1: Find the percent of a number

- "What is 30% of 80?"
- $0.30 \times 80 = 24$

Type 2: Find what percent

- "12 is what percent of 48?"
- $\frac{12}{48} = 0.25 = 25\%$

Type 3: Find the whole

- "15 is 30% of what number?"
- $0.30 \times x = 15, x = 50$

Percent Change:

$$\text{Percent change} = \frac{\text{new} - \text{old}}{\text{old}} \times 100\%$$

Discount/Markup Shortcuts:

- 20% off → multiply by 0.80
- 15% markup → multiply by 1.15
- Multiple discounts: apply one at a time (not additive!)

Guide 4: Data Displays—When to Use Mean vs. Median

Mean (Average):

- **Use when:** Data is fairly symmetric, no outliers
- **Advantage:** Uses all data points
- **Example:** Test scores: 85, 88, 90, 92, 95 → Mean = 90

Median (Middle Value):

- **Use when:** Data has outliers or is skewed
- **Advantage:** Not affected by extreme values
- **Example:** Salaries: \$30k, \$32k, \$35k, \$38k, \$200k → Median = \$35k (better than mean of \$67k)

Mode (Most Frequent):

- **Use when:** You want to know the most common value
- **Example:** Shoe sizes in a store (helpful for stocking)

Visual Guide:

Symmetric data (use mean):

```

      *
    * * *
  * * * * *
  -----
  
```

Skewed data with outlier (use median):

```

      *
    * *
  * * * *      *
  -----      (outlier)
  
```

Mini-Mistake Bank:

Mistake 1: "Mean is always best"

Correction: Not when there are outliers! Use median.

Mistake 2: Confusing range with interquartile range

Correction: Range = max - min. IQR = Q3 - Q1

Continue

- Q1 (more resistant to outliers)

Mistake 3: Calculating mean of percentages incorrectly

Correction: Can't just average percents—need to consider the whole amounts first.

8. Reference Sheets

Formula & Vocabulary Sheet (1-page)

RATIOS & PROPORTIONS

- **Unit Rate:** $\frac{\text{quantity}}{1 \text{ unit}}$
 - **Percent:** $\frac{\text{part}}{\text{whole}} \times 100\%$
 - **Proportional Relationship:** $y = kx$ (where k is constant of proportionality)
-

NUMBER OPERATIONS

Integer Rules:

- Same signs: Add, keep sign
- Different signs: Subtract, take larger's sign
- Multiply/Divide: Same signs $\rightarrow +$, Different $\rightarrow -$

Fractions:

- Add/Subtract: Common denominator needed
 - Multiply: $\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$
 - Divide: $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$
-

EXPRESSIONS & EQUATIONS

- **Distributive Property:** $a(b + c) = ab + ac$
- **Two-Step Equations:** Undo addition/subtraction first, then multiplication/division

- **Order of Operations:** PEMDAS (Parentheses, Exponents, Multiply/Divide left to right, Add/Subtract left to right)
-

GEOMETRY

2D Shapes:

- Circle Circumference: $C = 2\pi r$ or $C = \pi d$
- Circle Area: $A = \pi r^2$
- Triangle Area: $A = \frac{1}{2}bh$

3D Shapes:

- Rectangular Prism Volume: $V = lwh$
- Prism Volume: $V = Bh$ (B = base area)
- Cylinder Volume: $V = \pi r^2 h$
- Pyramid Volume: $V = \frac{1}{3}Bh$
- Surface Area: Add area of all faces

Angles:

- Complementary: Sum = 90°
 - Supplementary: Sum = 180°
 - Triangle angles: Sum = 180°
-

STATISTICS & PROBABILITY

- **Mean:** $\frac{\text{sum of values}}{\text{number of values}}$
- **Median:** Middle value (order data first)
- **Mode:** Most frequent value
- **Range:** Maximum – Minimum
- **Probability:** $P(\text{event}) = \frac{\text{favorable outcomes}}{\text{total outcomes}}$

- **Independent Events:** $P(A \text{ and } B) = P(A) \times P(B)$

Visual Aids

Percent Bar:



Example: Find 40% of 50

40% position $\rightarrow 0.40 \times 50 = 20$

Double Number Line (for Proportions):

Apples: 0 3 6 9 12

Cost: |----|----|----|----|
0 6 12 18 24

Shows: 3 apples = \$6, so 9 apples = \$18

Equation-Solving Flowchart:

Start with equation: $2x + 5 = 13$

↓

Is there addition/subtraction?

↓ YES

Undo it (subtract 5 from both sides)

↓

Result: $2x = 8$

↓

Is there multiplication/division?

↓ YES

Undo it (divide both sides by 2)

↓

Solution: $x = 4$

↓

CHECK by substituting back!

9. Study Schedule (4-Week Plan)

Time commitment: 30–45 minutes/day, 4 days/week

Week 1: Foundation & Assessment

Day	Focus	Activity	Time
Mon	Diagnostic	Complete diagnostic test	30 min
Tue	Topic A	Lesson A1 (Unit Rates)	40 min
Wed	Topic A	Practice Set (Foundation or On-Level based on diagnostic)	35 min
Thu	Topic B	Lesson B1 (Adding/Subtracting Integers)	40 min

Parent Note: Review diagnostic results. Focus on the priority topic identified.

Week 2: Building Skills

Day	Focus	Activity	Time
Mon	Topic A	Lesson A2 (Percent Problems)	40 min
Tue	Topic B	Lesson B2 (Multiplying/Dividing Integers) + Practice	45 min
Wed	Application	Project A2 (Budget Shopping)	45 min
Thu	Review	Mixed practice from Topics A & B (10 problems)	30 min

Week 3: Expanding Understanding

Day	Focus	Activity	Time
Mon	Topic C	Lesson C1 (Simplifying Expressions)	35 min
Tue	Topic C	Lesson C2 (Two-Step Equations) + Practice	45 min
Wed	Topic D	Lesson D1 (Circles) + Quick Practice	40 min
Thu	Mastery Check	Week 3 Mastery Check (8 problems)	30 min

Week 4: Application & Review

Day	Focus	Activity	Time
Mon	Topic E	Lesson E1 (Measures of Center) + Practice	40 min
Tue	Project	Choose one project to complete (D1 or E1)	45 min
Wed	Spiral Review	Mixed problems from all topics (12 problems)	35 min
Thu	Cumulative	Cumulative Mastery Check + Error Analysis	45 min

Light Math Games Options (for breaks or bonus days):

- KenKen puzzles
- Set card game
- 24 game
- Logic grid puzzles
- Sudoku

10. Recommended Resources

Note: Resources listed by description only; search online to access

Concept Explainer Videos

Resource 1: Khan Academy Grade 7 Math

- **Best for:** Clear video lessons on all Grade 7 topics
- **How to use:** Watch 5–10 minute videos before or after lessons for reinforcement
- **Parent tip:** Have student take notes while watching

Resource 2: Math Antics (YouTube Channel)

- **Best for:** Visual, animated explanations of basic concepts
- **How to use:** Great for students who struggle with reading-heavy materials
- **Parent tip:** Watch together and pause to discuss

Resource 3: Professor Dave Explains (YouTube Channel)

- **Best for:** Quick, engaging reviews of specific topics
 - **How to use:** 5-minute refreshers before practice sets
 - **Parent tip:** Good for "I forgot how to do this" moments
-

Interactive Practice Sites

Resource 4: IXL Math Grade 7

- **Best for:** Adaptive practice with immediate feedback
- **How to use:** 15–20 minutes after lessons for reinforcement
- **Parent tip:** Track which skills reach 80%+ mastery

Resource 5: DeltaMath

- **Best for:** Customizable problem sets with step-by-step help
- **How to use:** Teachers or parents can create assignments
- **Parent tip:** Use "homework help" mode for extra scaffolding

Resource 6: Desmos Activities

- **Best for:** Interactive, visual explorations (graphing, proportions)
 - **How to use:** Use specific activities like "Marble Slides" for functions
 - **Parent tip:** Very engaging; great for visual learners
-

Printable Worksheets & Problem Banks

Resource 7: Math-Aids.com

- **Best for:** Infinite customizable worksheets with answer keys
- **How to use:** Generate fresh practice problems as needed
- **Parent tip:** Print 2–3 sheets per topic for extra practice

Resource 8: Kuta Software

- **Best for:** High-quality, standardized practice sheets
 - **How to use:** Download PDF worksheets for each topic
 - **Parent tip:** Answers are included; good for independent work
-

Enrichment & Challenge

Resource 9: Art of Problem Solving (AoPS) Alcumus

- **Best for:** Advanced students seeking challenge problems
- **How to use:** 20–30 minutes, 2–3 times/week for enrichment
- **Parent tip:** Adjust difficulty level as student progresses

Resource 10: Math Counts

- **Best for:** Competition-style problems and puzzles
- **How to use:** Use past competition problems for challenge practice
- **Parent tip:** Great for students interested in math competitions

Resource 11: NRICH (Cambridge University)

- **Best for:** Open-ended, investigative math problems
 - **How to use:** Choose problems by topic; work through together
 - **Parent tip:** Focuses on problem-solving and reasoning
-

11. Accessibility & Differentiation

For Struggling Learners

Scaffolding Strategies:

1. Concrete → Visual → Abstract Progression

- Use manipulatives first (blocks for fractions, number lines for integers)
- Move to pictures and diagrams
- Finally introduce symbols and algorithms

2. Chunking

- Break lessons into 10–15 minute segments
- Take 2–3 minute breaks between chunks
- Limit practice sets to 5 problems at a time

3. Worked Examples

- Show 2–3 fully worked examples before student tries
- Use different colors for different steps
- Have student explain each step back to you

4. Sentence Frames for Word Problems

- "Let ____ = ____"
- "The equation is ____"
- "To solve, I first ____, then ____"

5. Error Analysis

- Show intentional mistakes, have student find and correct
- Builds metacognition and confidence

For Advanced Learners

Extension Strategies:

1. Open-Ended Tasks

- "Create your own problem where the answer is 24"
- "How many ways can you solve this?"
- "What happens if we change this constraint?"

2. Non-Routine Problems

- Logic puzzles requiring multiple steps
- Problems with multiple possible solutions
- "Prove why this works" challenges

3. Cross-Topic Connections

- "How is simplifying fractions like solving proportions?"
- "Connect probability to fractions and percent"

4. Real-World Depth

- Research projects (e.g., "How do architects use scale?")
- Multi-step applications with authentic data

5. Challenge Set Focus

- Always assign Challenge practice sets
- Introduce Algebra 1 concepts as preview (solving systems, exponent rules)

For Students with ADHD / Executive Function Challenges

Support Strategies:

1. Timers & Time Management

- Use visual timers (5, 10, 15-minute intervals)
- "Beat the clock" challenges for engagement
- Break 30-min sessions into 3×10 -min with movement breaks

2. Checklists

- Print daily checklist: ☐ Read problem ☐ Show work ☐ Check answer
- Use completion checklists for each practice set
- Visual progress tracker (stickers, checkmarks)

3. Minimize Distractions

- Clear workspace (only pencil, paper, calculator)
- Use noise-canceling headphones or white noise
- Designated "math spot" away from screens

4. Movement Breaks

- Jump 10 times between problems
- Wall push-ups or stretches every 10 minutes
- Walk while reciting times tables or formulas

5. Immediate Feedback

- Check answers after every 2–3 problems (not at the end)
- Celebrate small wins frequently
- Use online tools with instant feedback when possible

12. Assessment & Progress Tracking

Student Progress Tracker

Student Name: _____ Week: _____

Date	Topic	Activity	Problems Completed	Accuracy %	Time (min)	Notes / Reflections

Weekly Goals:

1. _____
2. _____
3. _____

Celebrate Success: What went well this week?

Areas to Work On: What needs more practice?

Mastery Criteria

"Ready to Move On" indicators:

- Accuracy of 80%+ on On-Level practice sets
- Can explain reasoning behind answers
- Completes problems independently with minimal prompts
- Mastery Check score of 7+ out of 10

"Needs Review" indicators:

- Accuracy below 70%
- Frequent need for hints or re-explanation
- Cannot connect concepts to real-world applications
- Mastery Check score below 5

Action Steps for "Needs Review":

1. Revisit lesson videos or explanations
2. Complete Foundation practice set (even if started at On-Level)
3. Work through problems together with verbal reasoning
4. Try a different representation (visual, concrete)
5. Re-test after 2–3 days of targeted practice

Self-Assessment Rubric (for students)

After completing each topic, rate yourself:

Skill	I can do this easily! (3)	I can do this with some help (2)	I need more practice (1)
Understand the concept			
Solve basic problems			
Solve word problems			
Explain my thinking			
Connect to real life			

My goal for next week: _____

Summary & Final Tips

For Parents:

- ✓ **Start with the diagnostic** to target the right level
- ✓ **Follow the 4-day/week schedule**—consistency builds skills
- ✓ **Mix it up:** lessons, practice, projects, and review
- ✓ **Celebrate progress, not just perfection**
- ✓ **Check understanding by having your child explain concepts**
- ✓ **Use real-world connections** (cooking, shopping, sports stats)
- ✓ **Adjust pacing**—some topics may take longer, and that's OK
- ✓ **Keep a positive mindset:** "This is challenging, AND I can learn it"

For Students:

- 🎯 **Do your best, learn from mistakes**
- 🎯 **Show your work**—it helps you AND helps find errors
- 🎯 **Ask "why," not just "what's the answer"**
- 🎯 **Practice a little every day beats cramming**
- 🎯 **Math is everywhere**—look for it in your world!

You have everything you need. Let's get started!

END OF GRADE 7 MATHEMATICS RESOURCE KIT